Math 105: Calculus with Algebra
Instructor: David Zureick-Brown ("DZB")

Worksheet archive

Gradescope code: B5J8EN

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Gradescope Instructions for submitting work in Math 105

You will be using the online Gradescope progam to submit your homework and exams. These instructions tell you how to sign up initially, and how to submit your written work.

Signing up for Gradescope the first time.

If you haven't used Gradescope for an **Amherst College** course before:

- Go to `http://www.gradescope.com`, click on "Sign up for free" (which may auto-scroll you to the bottom of the page), and select Sign up as [a] "Student".
- In the signup box:
  - Use the course entry code **B5J8EN**
  - Use your full name
  - Use your **Amherst College email** address. Or, if you are a Five-College student, use your email address from your own school.
  - Leave the "Student ID" entry blank.
- You will probably get an email asking to set a password for your account, so check your `amherst.edu` email inbox. (Or your email inbox through your own school, for Five-College students.)

Adding Math 105 to Gradescope.

If you **have** used Gradescope for an Amherst course before, and so you already have an account through your `amherst.edu` email, you still need to add Math 105, so:

- Go to `http://www.gradescope.com` and log in.
- Go to your Account Dashboard (click the Gradescope logo at upper left), and click "Add Course" at bottom right.
- Use the course code **B5J8EN**

(submission instructions on next page)

Submitting written work

First write it out on paper as you would normally. Then **scan it** to create a PDF. One method for scanning is the smartphone app **DropBox**. It makes nice clear scans, and it saves them directly into a folder so that you can have all your assignments in one place. **CamScanner** is another free scanning App, and there are others, too. **Gradescope** now has its own scanning app. You can also use a printer/scanner if you prefer.

Please be kind to our dear graders and make sure your submission is legible !

In particular, please leave some spacing between separate problems.

If you have a tablet computer, you may write your work there (instead of on paper) and save it as a PDF.

Some of you may know the math formatting package LaTeX and may want to use it in Math 105. That's fine, too; if so, you may write up your work in LaTeX and save the resulting PDF.

In short, any method is fine as long as it creates a legible **PDF** file and **NOT a photo**.

For example, if you use the DropBox app, then in your created *Math 105 Homework* Dropbox folder, you can select create (+) at the bottom of the screen and click the *Scan Document* option. Snap a shot of the first page of your homework, and then click [+] to snap shots of any subsequent pages. Do **not** use the *Take Photo* option.

After you have scanned/saved your work as a PDF, submit it on Gradescope as follows:

- Go to `http://www.gradescope.com` and log in.
- Select the course "Math 105, Fall 2026" and the appropriate assignment.
- Select "submit pdf" to submit your work in PDF format. Browse to find your PDF and upload.
- Now it is time to **tag** your problems. This is an **important step**, where you are telling Gradescope which problems are on which page(s).

  For each problem, select the pages of your submission where your written solution appears.

  I think the easiest thing to do is to click on the page of **your** homework upload where you wrote the given problem, and then click on the assigned problem listed. Repeat for each problem.

  You must tag the problems or else you will not get credit for your work.

  Gradescope will give you a warning when you go to submit your assignment if you have not selected the pages correctly. If you tag a problem incorrectly, you can fix it by clicking "More" and "Reselect Pages".
- Click Save or Submit.

After your assignment is graded, you will be able to see your score on the written problems, along with comments, on Gradescope. You should receive an email notifying you when each homework set is graded.

Math 105, Fall 2026, Lab Worksheet 1

1. Which of the following are true and which are false? If it is false, then explain briefly why.

   (a) $\dfrac{0}{1} = 0$ (b) $\dfrac{1}{0} = 0$ (c) $\dfrac{0}{0} = 0$

2. Simplify each of the following expressions. Show your work.

   (a) $\dfrac{\left(\frac{p}{q}\right)}{\left(\frac{s}{t}\right)}$ (b) $\dfrac{1}{\left(\frac{s}{t}\right)}$

   (c) $\dfrac{\left(\frac{p}{q}\right)}{s}$ (d) $\dfrac{p}{\left(\frac{s}{t}\right)}$

3. Write as a single fraction:

$$1 + \frac{1}{1+\frac{1}{a}}$$

   Hint: start with the fraction in the denominator.

4. Write as a single fraction:

$$\frac{\frac{1}{1-a} - \frac{1}{1-b}}{a-b}$$

5. Solve each of the following equations (if possible):

   (a) $x^2 - 4x - 21 = 0$ (b) $x^2 - x + 7 = 0$ (c) $x^2 + 2x - 4 = 0$

6. YES or NO: Does $\sqrt{x^2+4} = x + 2$? Why or why not?

   Recall: The Absolute Value Function $f(x) = |x|$ is a *piecewise defined function* defined by

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

7. Express $|2x+1|$ using piecewise notation. In other words, fill in the blanks:

$$|2x+1| = \begin{cases} _____ & \text{if } x \geq _____, \\ _____ & \text{if } x < _____. \end{cases}$$

8. Solve each inequality for x (i.e. find all x so that the inequality is true). Express your answer in terms of intervals and illustrate the solution set on the real number line. (Hint for part b: Think about distance between x and 5!)

   (a) $\left|\frac{x}{3} - 1\right| - 7 \geq -5$

   (b) $0 < |x - 5| < \frac{1}{2}$

   (c) $|x - 5|(2 - x) > 0$

9. Evaluate the absolute value in each of the following:

   (a) $|4 - x|$ if $x < 4$

   (b) $|4 - x|$ if $x > 4$

   By "evaluate", we mean: "what are all of the possible values of $|4 - x|$?"

10. Solve the following equations for x:

    (a) $|3x + 1| = 5$

    (b) $\left|\frac{2x - 1}{x + 1}\right| = 3$

Each person turn in your own solutions, using Gradescope, by 11:55pm tonight (Thursday).

Math 105, Fall 2026, Lab Worksheet 1

1. Which of the following are true and which are false?

   (a) **True.**
   $$\frac{0}{1} = 0.$$

   (b) **False.** The expression $\frac{1}{0}$ is undefined.

   (c) **False.** The expression $\frac{0}{0}$ is undefined.

2. Simplify each of the following expressions.

   (a)
   $$\frac{\frac{p}{q}}{\frac{s}{t}} = \frac{p}{q} \cdot \frac{t}{s} = \boxed{\frac{pt}{qs}}.$$

   (b)
   $$\frac{1}{\frac{s}{t}} = \boxed{\frac{t}{s}}.$$

   (c)
   $$\frac{\frac{p}{q}}{s} = \frac{p}{q} \cdot \frac{1}{s} = \boxed{\frac{p}{qs}}.$$

   (d)
   $$\frac{p}{\frac{s}{t}} = p \cdot \frac{t}{s} = \boxed{\frac{pt}{s}}.$$

3. Write as a single fraction:
   $$1 + \frac{1}{1 + \frac{1}{a}}.$$

   First,
   $$1 + \frac{1}{a} = \frac{a+1}{a}.$$

   Thus
   $$1 + \frac{1}{\frac{a+1}{a}} = 1 + \frac{a}{a+1} = \boxed{\frac{2a+1}{a+1}}.$$

4. Write as a single fraction:
   $$\frac{\frac{1}{1-a} - \frac{1}{1-b}}{a-b}.$$

   The numerator is
   $$\frac{1}{1-a} - \frac{1}{1-b} = \frac{(1-b)-(1-a)}{(1-a)(1-b)} = \frac{a-b}{(1-a)(1-b)}.$$

Therefore

$$\frac{\frac{a-b}{(1-a)(1-b)}}{a-b} = \boxed{\frac{1}{(1-a)(1-b)}}.$$

The original expression requires $a \neq 1$, $b \neq 1$, and $a \neq b$.

5. Solve each of the following equations.

(a)

$$x^2 - 4x - 21 = 0.$$

Factoring,

$$(x-7)(x+3) = 0,$$

so

$$\boxed{x = 7, -3}.$$

(b)

$$x^2 - x + 7 = 0.$$

The discriminant is

$$(-1)^2 - 4(1)(7) = 1 - 28 = -27 < 0.$$

Thus there are

$$\boxed{\text{no real solutions}}.$$

(c)

$$x^2 + 2x - 4 = 0.$$

Using the quadratic formula,

$$x = \frac{-2 \pm \sqrt{4+16}}{2} = \frac{-2 \pm \sqrt{20}}{2} = \boxed{-1 \pm \sqrt{5}}.$$

6. **NO.**

In general,

$$\sqrt{x^2+4} \neq x + 2.$$

For example, if $x = 1$, then

$$\sqrt{x^2+4} = \sqrt{5}, \qquad x + 2 = 3,$$

which are not equal.

Grading note: A single valid counterexample is enough. Students may also say that square roots do not distribute over addition. In fact, the equation itself is true only when $x = 0$.

7. We have

$$2x+1 \geq 0 \iff x \geq -\frac{1}{2}.$$

Therefore

$$\boxed{|2x+1| = \begin{cases} 2x+1 & \text{if } x \geq -\frac{1}{2}, \\ -2x-1 & \text{if } x < -\frac{1}{2}. \end{cases}}$$

8. Solve each inequality.

(a)

$$\left|\frac{x}{3} - 1\right| - 7 \geq -5.$$

Thus

$$\left|\frac{x}{3} - 1\right| \geq 2.$$

Hence

$$\frac{x}{3} - 1 \geq 2 \qquad \text{or} \qquad \frac{x}{3} - 1 \leq -2.$$

So

$$x \geq 9 \qquad \text{or} \qquad x \leq -3.$$

Therefore

$$\boxed{(-\infty, -3] \cup [9, \infty)}.$$

(b)

$$0 < |x-5| < \frac{1}{2}.$$

The inequality $|x-5| < \frac{1}{2}$ gives

$$\frac{9}{2} < x < \frac{11}{2},$$

while $|x-5| > 0$ excludes $x = 5$. Thus

$$\boxed{\left(\frac{9}{2}, 5\right) \cup \left(5, \frac{11}{2}\right)}.$$

(c)

$$|x-5|(2-x) > 0.$$

Since $|x-5| \geq 0$, the product is positive precisely when

$$|x-5| > 0 \quad \text{and} \quad 2-x > 0.$$

The second condition gives $x < 2$, which automatically makes $x \neq 5$. Therefore

$$\boxed{(-\infty, 2)}.$$

Grading note: For each part, students are also asked to draw the corresponding set on a number line.

9. Evaluate the absolute value.

   (a) If $x < 4$, then $4 - x > 0$, so

   $$\boxed{|4 - x| = 4 - x}.$$

   (b) If $x > 4$, then $4 - x < 0$, so

   $$\boxed{|4 - x| = x - 4}.$$

   In either case, as x ranges over the given domain, the possible values of $|4 - x|$ are all positive real numbers, i.e. $(0, \infty)$.

10. Solve the following equations.

   (a)

   $$|3x + 1| = 5.$$

   Thus

   $$3x + 1 = 5 \qquad \text{or} \qquad 3x + 1 = -5.$$

   Therefore

   $$\boxed{x = \frac{4}{3}, -2}.$$

   (b)

   $$\left|\frac{2x - 1}{x + 1}\right| = 3.$$

   Thus

   $$\frac{2x - 1}{x + 1} = 3 \qquad \text{or} \qquad \frac{2x - 1}{x + 1} = -3.$$

   In the first case,

   $$2x - 1 = 3x + 3,$$

   so

   $$x = -4.$$

   In the second case,

   $$2x - 1 = -3x - 3,$$

   so

   $$5x = -2, \qquad x = -\frac{2}{5}.$$

   Both are in the domain $x \neq -1$, so

   $$\boxed{x = -4, -\frac{2}{5}}.$$

Math 105, Fall 2026, Lab Worksheet 2

1. Let $g(x) = \dfrac{\dfrac{x}{x-1} - \dfrac{x+2}{x}}{x-2}$. Simplify $g(x)$. What is the domain of $g(x)$?

2. Find the domain of the following functions.

   (a) $f(x) = \dfrac{2}{(x+3)(x-5)}$

   (b) $f(x) = \dfrac{2}{x^2+2x-3}$

3. Find the equation of the line that is perpendicular to the line $x = 2$ and passes through $(1, 3)$.

4. Find the slope-intercept equation of the line L that passes through the point $(3, -1)$ and is perpendicular to the line $2x + 5y = 6$. THEN, does this new line L pass through $(1, -6)$?

5. Simplify the following expression: $\dfrac{x^2+6x+8}{x^2-4}$

6. Simplify the following expression: $\dfrac{1}{x} - \dfrac{1}{|x|}$

   Hint: you will need two cases here. Write out the piecewise definition of $|x|$.

7. Let $f(x) = \dfrac{1}{x+1}$. Compute and simplify $\dfrac{f(x+h) - f(x)}{h}$.

   Warning: $f(x+h) \neq f(x) + h$ **!!!**

8. Consider the function defined piecewise by $f(x) = \begin{cases} 2 - x & \text{if } x < -1 \\ x + 4 & \text{if } -1 < x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$

   Graph $f(x)$ carefully and find its Domain and Range.

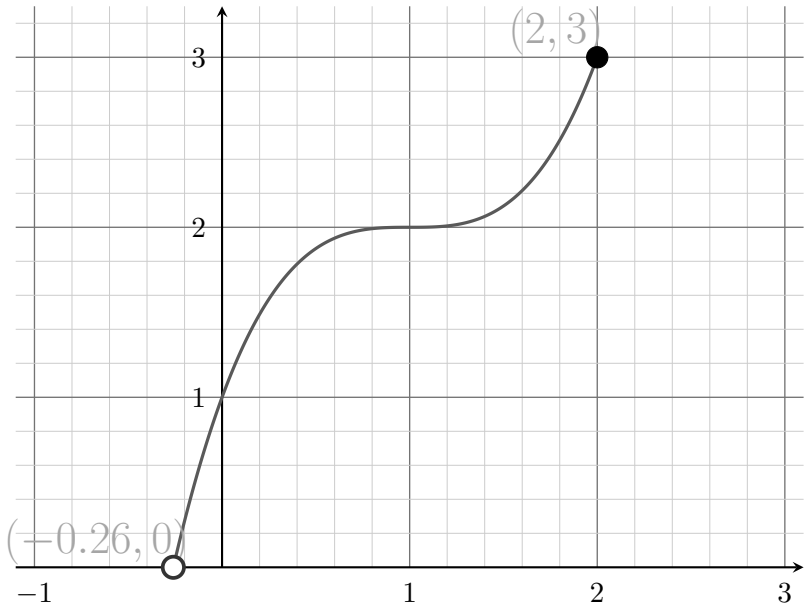
9. Consider the function $f(x) = x^2 - 6x - 7$. Compute and **simplify** each of the following. **(Careful about problem numbering when you submit your solutions.)**

   (a) $f(0)$ (b) $f(1)$ (c) $f(-3)$

   (d) $f(a)$ (e) $f(a + h)$ (f) $f(a + 2b)$

   (g) $f(-x)$ (h) $f(p^3)$ (i) $f(f(x))$

10. The picture below is the graph of a function $y = f(x)$.



    Answer the following:

    (a) Domain = ? (b) Range = ? (c) $f(0) = ?$

    (d) $f(-.26) = ?$ (e) $f(?) = 2$ (f) $f(1) + f(2) = ?$

Each person turn in your own solutions, using Gradescope, by 11:55pm tonight (Thursday).

Math 105, Fall 2026, Lab Worksheet 2 — Solution Key

1. Let
$$g(x) = \frac{\dfrac{x}{x-1} - \dfrac{x+2}{x}}{x-2}.$$

Solution. First simplify the numerator:
$$\begin{aligned}\frac{x}{x-1} - \frac{x+2}{x} &= \frac{x^2 - (x+2)(x-1)}{x(x-1)} \\ &= \frac{x^2 - (x^2+x-2)}{x(x-1)} \\ &= \frac{2-x}{x(x-1)} = -\frac{x-2}{x(x-1)}.\end{aligned}$$

Thus
$$g(x) = -\frac{1}{x(x-1)}.$$

The original expression is not defined at 0, 1, or 2, so the domain is
$$\boxed{\text{Dom}(g) = \mathbb{R} \setminus \{0,1,2\}}.$$

This notation means "All real numbers except for 0,1,2". One can also write this in interval notation.

2. Find the domain of the following functions.

(a) $f(x) = \dfrac{2}{(x+3)(x-5)}$

The denominator is zero at $x = -3$ and $x = 5$. Therefore
$$\boxed{\text{Dom}(f) = \mathbb{R} \setminus \{-3,5\}}.$$

(b) $f(x) = \dfrac{2}{x^2+2x-3}$

Factor the denominator:
$$x^2 + 2x - 3 = (x+3)(x-1).$$

Hence
$$\boxed{\text{Dom}(f) = \mathbb{R} \setminus \{-3,1\}}.$$

3. Find the equation of the line that is perpendicular to $x = 2$ and passes through $(1,3)$.

Solution. The line $x = 2$ is vertical, so any perpendicular line is horizontal. Through $(1,3)$ this gives
$$\boxed{y = 3}.$$

4. Find the slope-intercept equation of the line L that passes through $(3,-1)$ and is perpendicular to $2x + 5y = 6$. Then determine whether L passes through $(1,-6)$.

Solution. Rewrite the given line as
$$y = -\frac{2}{5}x + \frac{6}{5},$$

so its slope is $-\frac{2}{5}$. A perpendicular line has slope $\frac{5}{2}$. Using point-slope form through $(3,-1)$,

$$y+1=\frac{5}{2}(x-3),$$

so

$$\boxed{y=\frac{5}{2}x-\frac{17}{2}}.$$

At $x=1$,

$$y=\frac{5}{2}-\frac{17}{2}=-6,$$

so $\boxed{\text{yes, } L \text{ passes through } (1,-6)}$.

5. Simplify

$$\frac{x^2+6x+8}{x^2-4}.$$

Solution.

$$\frac{x^2+6x+8}{x^2-4}=\frac{(x+2)(x+4)}{(x+2)(x-2)}=\boxed{\frac{x+4}{x-2}}.$$

6. Simplify

$$\frac{1}{x}-\frac{1}{|x|}.$$

Solution. Recall

$$|x|=\begin{cases}x, & x\geq 0,\\ -x, & x<0.\end{cases}$$

Since the original expression is undefined at $x=0$, we only need the cases $x>0$ and $x<0$.

If $x>0$, then $|x|=x$, so

$$\frac{1}{x}-\frac{1}{|x|}=0.$$

If $x<0$, then $|x|=-x$, so

$$\frac{1}{x}-\frac{1}{|x|}=\frac{1}{x}-\frac{1}{-x}=\frac{2}{x}.$$

Therefore

$$\boxed{\frac{1}{x}-\frac{1}{|x|}=\begin{cases}\frac{2}{x}, & x<0,\\ 0, & x>0.\end{cases}}$$

7. Let $f(x)=\dfrac{1}{x+1}$. Compute and simplify

$$\frac{f(x+h)-f(x)}{h}.$$

Solution.

$$\begin{aligned}\frac{f(x+h)-f(x)}{h} &= \frac{\dfrac{1}{x+h+1}-\dfrac{1}{x+1}}{h}\\ &= \frac{(x+1)-(x+h+1)}{h(x+h+1)(x+1)}\\ &= \frac{-h}{h(x+h+1)(x+1)}\\ &= \boxed{-\frac{1}{(x+h+1)(x+1)}}.\end{aligned}$$

(For the original difference quotient, $h \neq 0$, $x \neq -1$, and $x+h \neq -1$.)

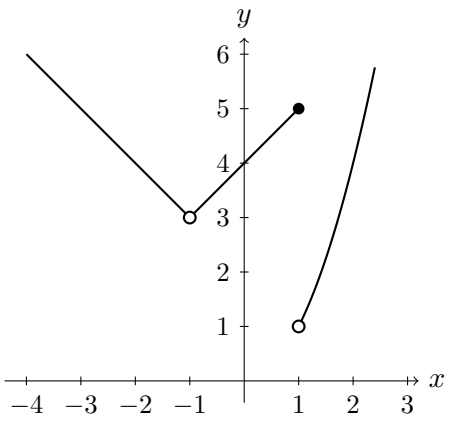
8. Consider

$$f(x) = \begin{cases} 2-x, & x < -1,\\ x+4, & -1 < x \leq 1,\\ x^2, & x > 1.\end{cases}$$

Graph f and find its domain and range.

Solution. The graph has:

- the line $y = 2 - x$ for $x < -1$, with an open circle at $(-1, 3)$;
- the line $y = x + 4$ for $-1 < x \leq 1$, with an open circle at $(-1, 3)$ and a closed circle at $(1, 5)$;
- the parabola $y = x^2$ for $x > 1$, with an open circle at $(1, 1)$.



The only missing input is $x = -1$, so

$$\boxed{\text{Dom}(f) = (-\infty, -1) \cup (-1, \infty)}.$$

The third piece x^2 for $x > 1$ already gives every output greater than 1, but not 1 itself. Thus

$$\boxed{\text{Range}(f) = (1, \infty)}.$$

9. Let $f(x) = x^2 - 6x - 7$. Compute and simplify each expression.

(a)
$$\boxed{f(0) = -7}.$$

(b)
$$\boxed{f(1) = 1 - 6 - 7 = -12}.$$

(c)
$$\boxed{f(-3) = 9 + 18 - 7 = 20}.$$

(d)
$$\boxed{f(a) = a^2 - 6a - 7}.$$

(e)
$$\begin{aligned} f(a+h) &= (a+h)^2 - 6(a+h) - 7 \\ &= \boxed{a^2 + 2ah + h^2 - 6a - 6h - 7}. \end{aligned}$$

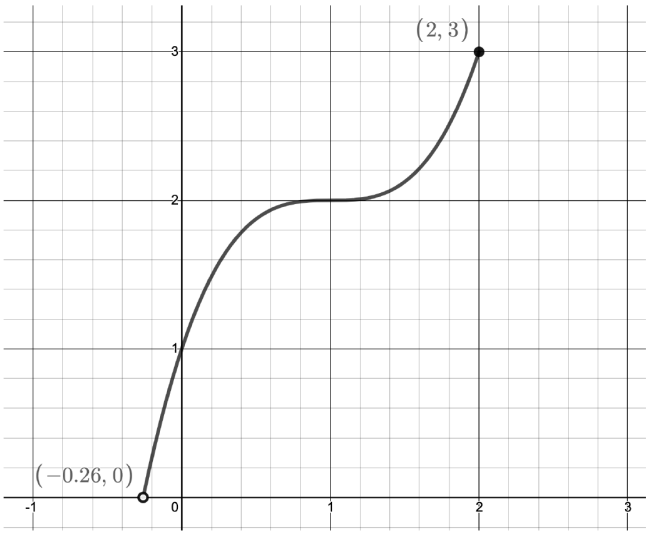
(f)
$$\begin{aligned} f(a+2b) &= (a+2b)^2 - 6(a+2b) - 7 \\ &= \boxed{a^2 + 4ab + 4b^2 - 6a - 12b - 7}. \end{aligned}$$

(g)
$$\boxed{f(-x) = x^2 + 6x - 7}.$$

(h)
$$\boxed{f(p^3) = p^6 - 6p^3 - 7}.$$

(i)
$$\begin{aligned} f(f(x)) &= (x^2 - 6x - 7)^2 - 6(x^2 - 6x - 7) - 7 \\ &= \boxed{x^4 - 12x^3 + 16x^2 + 120x + 84}. \end{aligned}$$

10. The picture below is the graph of a function $y = f(x)$.



Solution. The left endpoint $(-0.26, 0)$ is open and the right endpoint $(2, 3)$ is closed. Reading the marked grid points from the graph gives

(a) Domain $= (-0.26, 2]$,	(b) Range $= (0, 3]$,	(c) $f(0) = 1$,
(d) $f(-0.26)$ is undefined,	(e) $f(1) = 2$,	(f) $f(1) + f(2) = 2 + 3 = 5$.

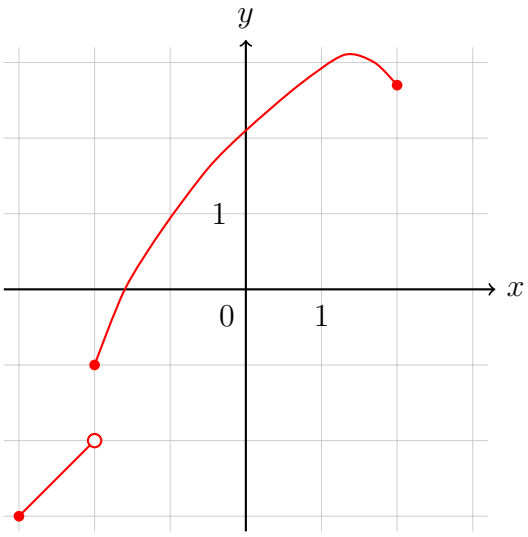
Math 105, Fall 2026, Worksheet 3

1. Graph the function $y = 1 - 2\sqrt{x-3}$ by hand, not by plotting points, but by starting with the graph of $\sqrt{x}$ and then applying the appropriate transformations.

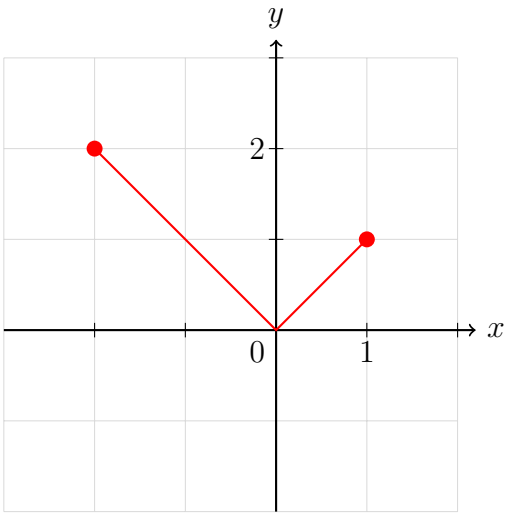
2. Graph the functions $F(x) = \sqrt{|x|}$ and $F(x) = \sqrt{2 + |x|}$

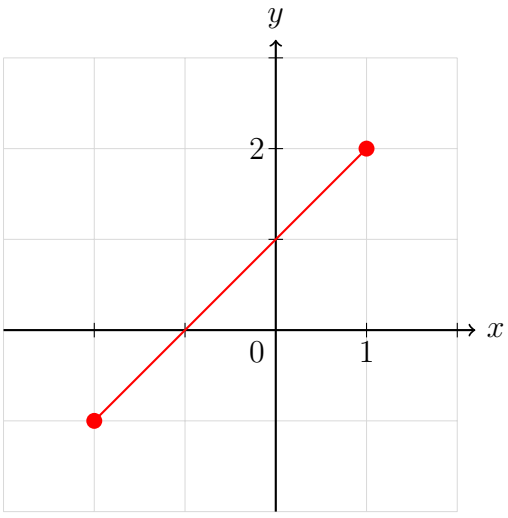
3. The graph of f is given below. Graph each of the following:

   (a) $y = f(x) - 3$ (b) $y = f(x + 1)$
   (c) $y = \frac{1}{2}f(x)$ (d) $y = -f(x)$



4. The graphs of f and g are given below. (f is the left one.) Graph $f + g$.





5. Let $f(x) = \frac{1}{2-x}$ and $g = \frac{1}{3x}$. Find formulas for each of the following (and simplify your formulas).

   (a) $f \circ g$ (b) $g \circ f$

   For each limit, either find the value (if it exists), or answer $+\infty$, $-\infty$, or "DNE" (for "Does Not Exist").

   Then, determine if the point (a, L) is on the graph of $y = f(x)$ or if it is a hole on the graph, where $L = \lim_{x \to a} f(x)$.

6. Find both $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$ and $\lim_{x \to 1} \dfrac{x^2 - 4}{x - 2}$. Why is one easier than the other?

7. $\lim_{x \to 2^-} \dfrac{x^2 + 6x + 8}{x - 2}$

8. $\lim_{x \to 2} \dfrac{x^2 + 5x - 14}{x^2 - 8x + 12}$

9. $\lim_{x \to 0} \dfrac{x + 1}{x(x + 2)}$

   Note: Try taking the two one-sided limits separately ($\lim_{x \to 0^-}$ and $\lim_{x \to 0^+}$). If you get different results, then answer "DNE" for the overall limit. You should try this strategy in situations where you need to do a "sign analysis" (when you see $c/0$) but aren't sure how to proceed for the two-sided limit.

10. $\lim_{x \to 3} \dfrac{\frac{x}{x-2} - \frac{x+6}{x}}{x - 3}$

Math 105, Fall 2026, Worksheet 3 Solutions

1. Graph

$$y = 1 - 2\sqrt{x-3}$$

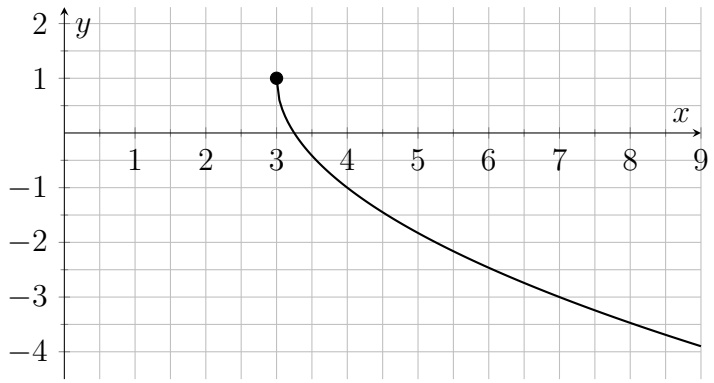
starting from $y = \sqrt{x}$.

Starting with $y = \sqrt{x}$:

1. shift the graph 3 units to the right;
2. stretch vertically by a factor of 2;
3. reflect across the x-axis;
4. shift up 1 unit.

The graph begins at $(3, 1)$ and then decreases as x increases. Its domain and range are

$$\boxed{[3, \infty)} \qquad \text{and} \qquad \boxed{(-\infty, 1]}.$$



2. Graph

$$F(x) = \sqrt{|x|} \qquad \text{and} \qquad G(x) = \sqrt{2 + |x|}.$$

For $F(x) = \sqrt{|x|}$, the absolute value makes the graph symmetric about the y-axis:

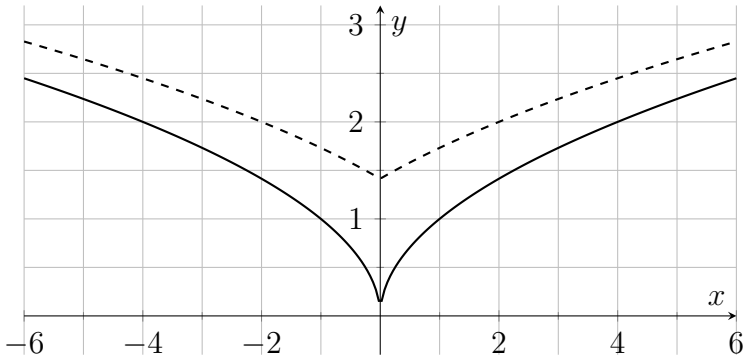
$$F(x) = \begin{cases} \sqrt{x}, & x \geq 0, \\ \sqrt{-x}, & x \leq 0. \end{cases}$$

Thus its domain is all real numbers and its range is $[0, \infty)$.

For $G(x) = \sqrt{2 + |x|}$, the graph is again symmetric about the y-axis. Its smallest value occurs at $x = 0$:

$$G(0) = \sqrt{2}.$$

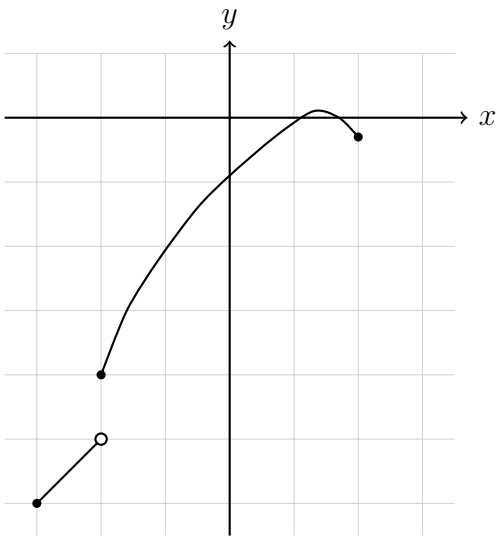
Thus its domain is all real numbers and its range is $[\sqrt{2}, \infty)$.



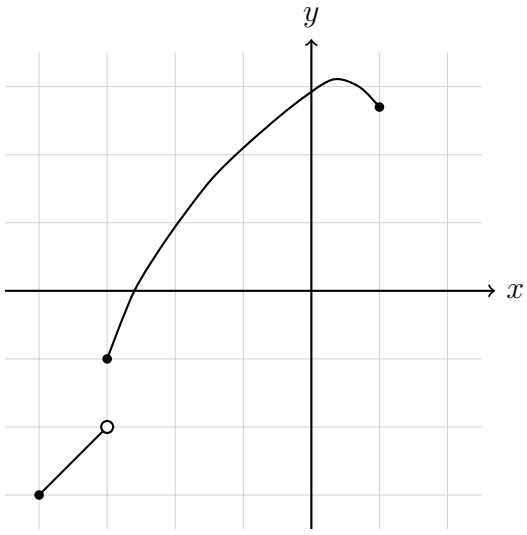
$\sqrt{|x|}$ - - - $\sqrt{2+|x|}$

3. The original graph of f is transformed in four different ways.

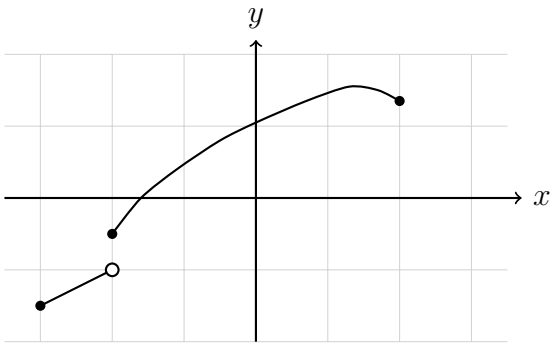
   (a) $y = f(x) - 3$ shifts the entire graph **down 3 units**.



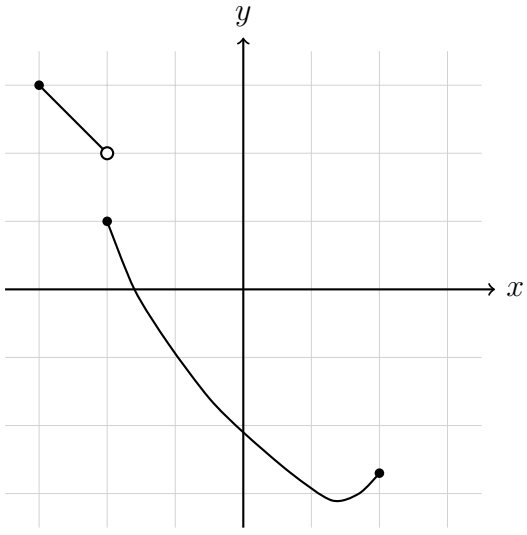
   (b) $y = f(x + 1)$ shifts the entire graph **left 1 unit**.



(c) $y = \frac{1}{2}f(x)$ compresses all vertical distances by a factor of $\frac{1}{2}$.



(d) $y = -f(x)$ reflects the graph across the x-axis.



4. The graph on the left is f and the graph on the right is g.

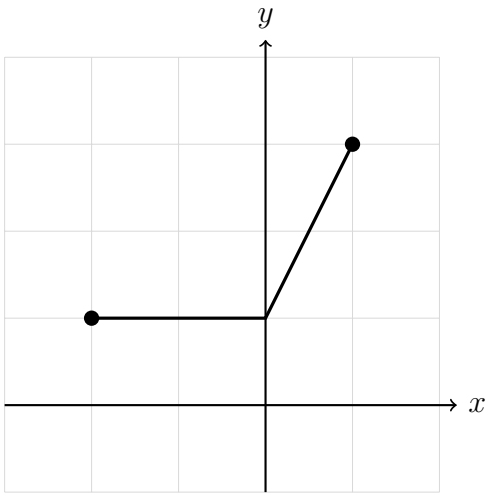
From the graphs,

$$f(x) = \begin{cases} -x, & -2 \le x \le 0, \\ x, & 0 \le x \le 1, \end{cases} \qquad g(x) = x + 1 \quad (-2 \le x \le 1).$$

Therefore

$$(f+g)(x) = \begin{cases} 1, & -2 \le x \le 0, \\ 2x+1, & 0 \le x \le 1. \end{cases}$$

So the sum is horizontal from $(-2, 1)$ to $(0, 1)$, then rises along the line $y = 2x + 1$ from $(0, 1)$ to $(1, 3)$.



5. Let

$$f(x) = \frac{1}{2-x}, \qquad g(x) = \frac{1}{3x}.$$

(a)

$$\begin{aligned} (f \circ g)(x) &= f\left(\frac{1}{3x}\right) \\ &= \frac{1}{2 - \frac{1}{3x}} \\ &= \frac{1}{\frac{6x-1}{3x}} \\ &= \boxed{\frac{3x}{6x-1}}. \end{aligned}$$

The composition requires

$$\boxed{x \neq 0, \qquad x \neq \frac{1}{6}}.$$

The restriction $x \neq 0$ is still necessary even though it disappears from the simplified formula, because $g(0)$ is undefined.

(b)

$$\begin{aligned}(g \circ f)(x) &= g\left(\frac{1}{2-x}\right) \\ &= \frac{1}{3\left(\frac{1}{2-x}\right)} \\ &= \boxed{\frac{2-x}{3}}.\end{aligned}$$

Here we must keep the restriction

$$\boxed{x \neq 2},$$

because $f(2)$ is undefined.

6. Find

$$\lim_{x\to 2} \frac{x^2-4}{x-2} \qquad \text{and} \qquad \lim_{x\to 1} \frac{x^2-4}{x-2}.$$

For the first limit, direct substitution gives $0/0$, so factor:

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2 \qquad (x \neq 2).$$

Therefore

$$\boxed{\lim_{x\to 2} \frac{x^2-4}{x-2} = 4}.$$

The original function is not defined at $x = 2$, so $(2, 4)$ is a **hole**.

For the second limit the denominator is nonzero at $x = 1$, so direct substitution works:

$$\boxed{\lim_{x\to 1} \frac{x^2-4}{x-2} = \frac{-3}{-1} = 3}.$$

The point $(1, 3)$ is actually **on the graph**.

The $x \to 1$ limit is easier because it can be found by direct substitution. The $x \to 2$ limit first gives the indeterminate form $0/0$ and must be simplified.

7.

$$\lim_{x\to 2^-} \frac{x^2+6x+8}{x-2}.$$

As $x \to 2^-$,

$$x^2 + 6x + 8 \to 24 > 0$$

while

$$x - 2 \to 0^-.$$

Thus a positive number is being divided by a very small negative number, so

$$\boxed{-\infty}.$$

There is a vertical asymptote at $x = 2$.

8.
$$\lim_{x\to 2}\frac{x^2+5x-14}{x^2-8x+12}.$$

Factor both numerator and denominator:

$$x^2+5x-14=(x+7)(x-2),$$

$$x^2-8x+12=(x-2)(x-6).$$

For $x\neq 2$,

$$\frac{x^2+5x-14}{x^2-8x+12}=\frac{x+7}{x-6}.$$

Therefore

$$\boxed{\lim_{x\to 2}\frac{x^2+5x-14}{x^2-8x+12}=\frac{9}{-4}=-\frac{9}{4}}.$$

Since the original function is undefined at $x=2$, the point

$$\boxed{\left(2,-\frac{9}{4}\right)}$$

is a hole.

9.
$$\lim_{x\to 0}\frac{x+1}{x(x+2)}.$$

Near $x=0$, both $x+1$ and $x+2$ are positive, so the sign is controlled by the factor x.

From the left,

$$\lim_{x\to 0^-}\frac{x+1}{x(x+2)}=-\infty.$$

From the right,

$$\lim_{x\to 0^+}\frac{x+1}{x(x+2)}=+\infty.$$

Since the two one-sided limits are different,

$$\boxed{\lim_{x\to 0}\frac{x+1}{x(x+2)}=\text{DNE}}.$$

There is a vertical asymptote at $x=0$.

10.
$$\lim_{x\to 3}\frac{\frac{x}{x-2}-\frac{x+6}{x}}{x-3}.$$

First combine the two fractions in the numerator:

$$\begin{aligned}\frac{x}{x-2}-\frac{x+6}{x} &= \frac{x^2-(x+6)(x-2)}{x(x-2)}\\ &= \frac{x^2-(x^2+4x-12)}{x(x-2)}\\ &= \frac{-4x+12}{x(x-2)}\\ &= \frac{-4(x-3)}{x(x-2)}.\end{aligned}$$

Thus, for $x \neq 3$,

$$\frac{\frac{x}{x-2}-\frac{x+6}{x}}{x-3} = \frac{-4}{x(x-2)}.$$

Therefore

$$\boxed{\lim_{x\to 3}\frac{\frac{x}{x-2}-\frac{x+6}{x}}{x-3} = -\frac{4}{3}}.$$

The original expression is undefined at $x = 3$, so the point

$$\boxed{\left(3, -\frac{4}{3}\right)}$$

is a hole.

Math 105, Fall 2026, Worksheet 4

1. Consider the function defined piecewise by $f(x) = \begin{cases} 2-x & \text{if } x < -1 \\ x+4 & \text{if } -1 < x \leq 1 \\ (x-2)^2 & \text{if } x > 1 \end{cases}$

   Graph $f(x)$ carefully and find its Domain and Range. Compute $\lim_{x\to 1^+} f(x)$. Compute $\lim_{x\to 1^-} f(x)$. Compute $\lim_{x\to 1} f(x)$, if it exists.

2. Consider the function defined by $f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4 \end{cases}$

   Graph $f(x)$ carefully and find its Domain and Range. Compute $\lim_{x\to 4} f(x)$, if it exists.

   Hint: Investigate that one-sided limits at 4.

3. Let $f(x) = \begin{cases} \sqrt{x-3}+1 & \text{if } x > 3 \\ 0 & \text{if } x = 3 \\ (x-2)^2 & \text{if } 1 < x < 3 \\ x + \dfrac{1}{3} & \text{if } 0 < x \leq 1 \\ \dfrac{1}{x+3} & \text{if } x < 0. \end{cases}$

   (a) Carefully sketch the graph of $f(x)$.

   (b) State the **Domain** of the function $f(x)$.

   Compute the following limits.

   (c) $\lim_{x\to -3} f(x)$

   (d) $\lim_{x\to 0} f(x)$

   (e) $\lim_{x\to 1} f(x)$

   (f) $\lim_{x\to 3} f(x)$

4. Given that

$$\lim_{x\to 3} f(x) = 9, \qquad \lim_{x\to 3} g(x) = -3, \qquad \lim_{x\to 3} h(x) = 0,$$

find the limits that can be determined from this information. If a limit cannot be determined, explain why.

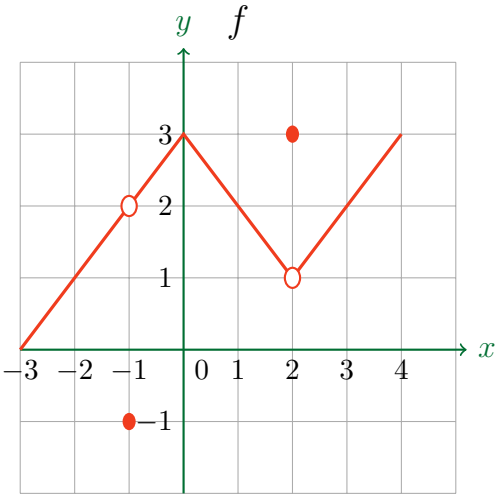
(a) $\lim_{x\to 3} (2f(x) + g(x))$

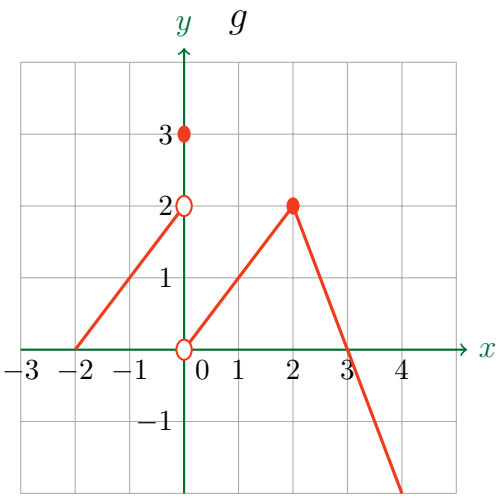
(b) $\lim_{x\to 3} (g(x))^2$

(c) $\lim_{x\to 3} \sqrt{f(x)}$

(d) $\lim_{x\to 3} \frac{f(x)}{h(x)}$

5. The graphs of f and g are given below. Use them to evaluate each limit, if it exists. If the limit does not exist, explain why.





(a) $\lim_{x\to 2} (f(x) + g(x))$

(b) $\lim_{x\to 0} (f(x)g(x))$

(c) $\lim_{x\to -1} (f(x)g(x))$

(d) $\lim_{x\to 2} (x^2 f(x))$

Compute the following limits (and show your work)

6. $\lim_{x\to 7} \dfrac{x^2 - 4x - 21}{x^2 - 3x}$

7. $\lim_{x\to 2} \dfrac{3 - \sqrt{x+7}}{x^2 - 3x + 2}$

8. $\lim_{x\to 4} \dfrac{x^2 - 3x - 4}{|x - 4|}$

9. $\lim_{x\to -5} \dfrac{\frac{1}{1-x} - \frac{1}{6}}{x^2 + 3x - 10}$

10. $\lim_{x\to 0} \dfrac{(x-1)^3}{x^4(x+2)}$

10. Let $f(x) = x^2$. Compute the difference quotient $\dfrac{f(x+h) - f(x)}{h}$.

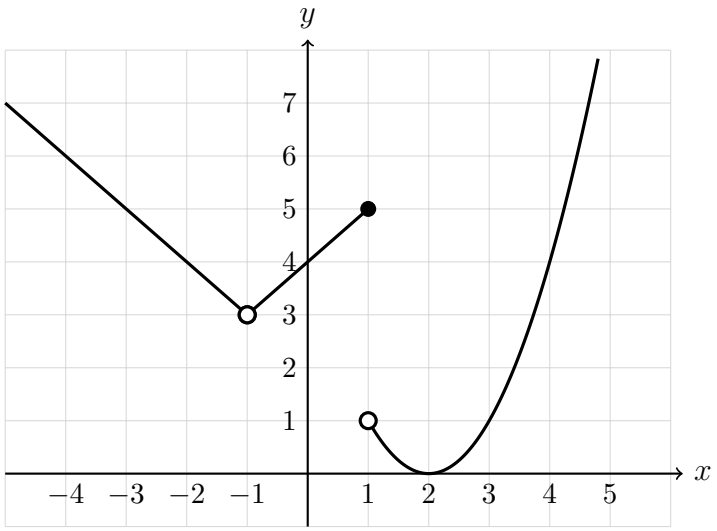
Simplify until you cancel the h in the denominator. Then compute $\lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$. (Note that we are treating h as the variable, instead of x.)

Math 105, Fall 2026, Worksheet 4 — Solutions

1. Consider

$$f(x) = \begin{cases} 2 - x & \text{if } x < -1, \\ x + 4 & \text{if } -1 < x \leq 1, \\ (x-2)^2 & \text{if } x > 1. \end{cases}$$

A sketch is shown below.



The function is defined for every real number except $x = -1$, so

$$\boxed{\text{Domain} = (-\infty, -1) \cup (-1, \infty).}$$

The piece $(x-2)^2$ for $x > 1$ takes every value in $[0, \infty)$, so

$$\boxed{\text{Range} = [0, \infty).}$$

As $x \to 1^+$, we use $(x-2)^2$, so

$$\boxed{\lim_{x \to 1^+} f(x) = 1.}$$

As $x \to 1^-$, we use $x + 4$, so

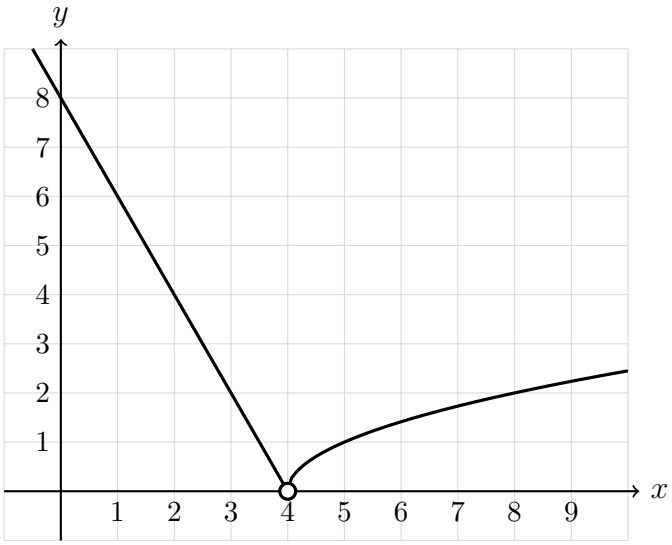
$$\boxed{\lim_{x \to 1^-} f(x) = 5.}$$

Since the one-sided limits are different,

$$\boxed{\lim_{x \to 1} f(x) \text{ does not exist.}}$$

2. Consider

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4, \\ 8 - 2x & \text{if } x < 4. \end{cases}$$



The only excluded input is $x = 4$, so

$$\boxed{\text{Domain} = (-\infty, 4) \cup (4, \infty).}$$

Both pieces take only positive values, and values can be arbitrarily close to 0, so

$$\boxed{\text{Range} = (0, \infty).}$$

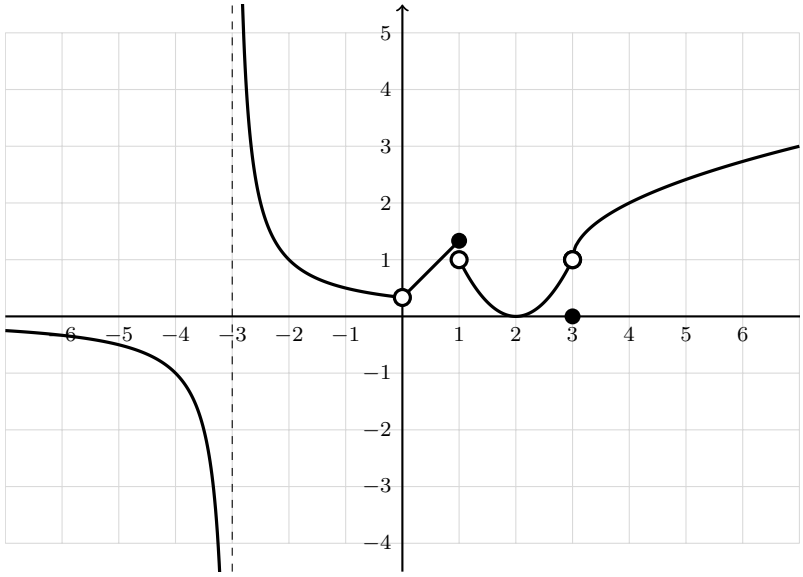
From the left, $8 - 2x \to 0$ as $x \to 4^-$. From the right, $\sqrt{x-4} \to 0$ as $x \to 4^+$. Therefore

$$\boxed{\lim_{x \to 4} f(x) = 0.}$$

3. Let

$$f(x) = \begin{cases} \sqrt{x-3} + 1 & \text{if } x > 3, \\ 0 & \text{if } x = 3, \\ (x-2)^2 & \text{if } 1 < x < 3, \\ x + \dfrac{1}{3} & \text{if } 0 < x \leq 1, \\ \dfrac{1}{x+3} & \text{if } x < 0. \end{cases}$$

(a) A sketch is shown below.



(b) The function is undefined at $x = -3$ and $x = 0$, and is defined everywhere else. Thus

$$\boxed{\text{Domain} = \mathbb{R} \setminus \{-3, 0\}.}$$

(c) Near $x = -3$, the relevant formula is $1/(x+3)$. As $x \to -3^-$, $f(x) \to -\infty$, while as $x \to -3^+$, $f(x) \to +\infty$. Therefore

$$\boxed{\lim_{x \to -3} f(x) \text{ does not exist.}}$$

(d) As $x \to 0^-$, $1/(x+3) \to 1/3$, and as $x \to 0^+$, $x + 1/3 \to 1/3$. Hence

$$\boxed{\lim_{x \to 0} f(x) = \frac{1}{3}.}$$

(e) As $x \to 1^-$, $x + 1/3 \to 4/3$, while as $x \to 1^+$, $(x-2)^2 \to 1$. Therefore

$$\boxed{\lim_{x \to 1} f(x) \text{ does not exist.}}$$

(f) As $x \to 3^-$, $(x-2)^2 \to 1$, and as $x \to 3^+$, $\sqrt{x-3} + 1 \to 1$. Therefore

$$\boxed{\lim_{x \to 3} f(x) = 1.}$$

Notice that $f(3) = 0$; the value of the function does not have to equal the limit.

4. Given that

$$\lim_{x\to 3} f(x) = 9, \qquad \lim_{x\to 3} g(x) = -3, \qquad \lim_{x\to 3} h(x) = 0,$$

find the limits that can be determined.

(a) By the sum and constant-multiple laws,

$$\lim_{x\to 3} (2f(x) + g(x)) = 2(9) + (-3) = \boxed{15}.$$

(b)

$$\lim_{x\to 3} (g(x))^2 = (-3)^2 = \boxed{9}.$$

(c) Since the square-root function is continuous at 9,

$$\lim_{x\to 3} \sqrt{f(x)} = \sqrt{9} = \boxed{3}.$$

(d) The quotient law cannot be used because the denominator approaches 0. Since $f(x) \to 9$, the numerator stays positive near $x = 3$, but the given information does not tell us from which side $h(x)$ approaches 0. The quotient could approach $+\infty$, $-\infty$, or fail to have a two-sided limit. Thus

$$\boxed{\text{the limit cannot be determined from the given information.}}$$

5. For the graphs of f and g:

(a) From the graphs, $f(x) \to 1$ and $g(x) \to 2$ as $x \to 2$. Therefore

$$\boxed{\lim_{x\to 2} (f(x) + g(x)) = 1 + 2 = 3.}$$

(b) As $x \to 0$, $f(x) \to 3$. However,

$$\lim_{x\to 0^-} g(x) = 2 \qquad \text{and} \qquad \lim_{x\to 0^+} g(x) = 0.$$

Thus the product approaches 6 from the left and 0 from the right, so

$$\boxed{\lim_{x\to 0} f(x)g(x) \text{ does not exist.}}$$

(c) As $x \to -1$, the graphs show $f(x) \to 2$ and $g(x) \to 1$. Hence

$$\boxed{\lim_{x\to -1} f(x)g(x) = 2.}$$

(d) As $x \to 2$, $x^2 \to 4$ and $f(x) \to 1$. Therefore

$$\boxed{\lim_{x\to 2} x^2 f(x) = 4.}$$

Compute the following limits.

6.
$$\lim_{x\to 7} \frac{x^2-4x-21}{x^2-3x}.$$
The denominator is nonzero at $x=7$, so direct substitution gives
$$\boxed{0}.$$

7.
$$\lim_{x\to 2} \frac{3-\sqrt{x+7}}{x^2-3x+2}.$$
Direct substitution gives $0/0$. Rationalize the numerator:
$$\frac{3-\sqrt{x+7}}{(x-1)(x-2)} \cdot \frac{3+\sqrt{x+7}}{3+\sqrt{x+7}} = \frac{9-(x+7)}{(x-1)(x-2)(3+\sqrt{x+7})}.$$
Since $9-(x+7) = 2-x = -(x-2)$, this becomes
$$-\frac{1}{(x-1)(3+\sqrt{x+7})}.$$
Therefore
$$\boxed{-\frac{1}{6}}.$$

8.
$$\lim_{x\to 4} \frac{x^2-3x-4}{|x-4|}.$$
Factor the numerator:
$$\frac{x^2-3x-4}{|x-4|} = (x+1)\frac{x-4}{|x-4|}.$$
From the right, this approaches 5; from the left, it approaches -5. Since the one-sided limits are different,
$$\boxed{\text{DNE.}}$$

9.
$$\lim_{x\to -5} \frac{\dfrac{1}{1-x}-\dfrac{1}{6}}{x^2+3x-10}.$$
First combine the numerator:
$$\frac{1}{1-x}-\frac{1}{6} = \frac{x+5}{6(1-x)}.$$

Also, $x^2 + 3x - 10 = (x + 5)(x - 2)$. After canceling $x + 5$,

$$\frac{1}{6(1 - x)(x - 2)}.$$

Substituting $x = -5$ gives

$$\boxed{-\frac{1}{252}}.$$

10.

$$\lim_{x \to 0} \frac{(x - 1)^3}{x^4(x + 2)}.$$

As $x \to 0$, the numerator approaches -1. Since $x^4 > 0$ on both sides of 0 and $x+2 > 0$ near 0, the denominator approaches 0 through positive values. Hence

$$\boxed{-\infty}.$$

10. Let $f(x) = x^2$. Then

$$\begin{aligned}
\frac{f(x + h) - f(x)}{h} &= \frac{(x + h)^2 - x^2}{h} \\
&= \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \frac{h(2x + h)}{h} \\
&= 2x + h.
\end{aligned}$$

Therefore

$$\boxed{\lim_{h \to 0} \frac{f(x + h) - f(x)}{h} = 2x.}$$

Instructor: David Zureick-Brown
Fall 2026

Amherst College
Dept. of Mathematics

Math 105 ∼ Calculus with Algebra: Midterm 1

- Do this exam during the first 55 minutes of the worksheet session.
- You are free to leave if you finish.
- But, after that I will stick around and answer any questions from the exam.
- Solutions are posted online.
- You can also ask about the practice exam in office hours.
- *Good luck!*

1. Consider the line L given by $x + 3y = 6$.
   (a) Sketch this line L.
   (b) Find the equation of the new line M that is **perpendicular** to the first line L, $x + 3y = 6$, and passes through the point $(1, -2)$.
   (c) Sketch this new line M found in (b).

2. Evaluate each of the following limits. Please justify your answers. Be clear if the limit equals a value, $+\infty$ or $-\infty$, or Does Not Exist.
   (a) $\lim_{x \to -7} \dfrac{x^2 + 5x - 14}{x^2 - 4x + 4} =$
   (b) $\lim_{x \to 4} \dfrac{x^2 - 9x + 20}{|4 - x|} =$
   (c) $\lim_{x \to 2} \dfrac{x^2 + 5x - 14}{x^2 - 4x + 4} =$
   (d) $\lim_{x \to 8} \dfrac{3 - \sqrt{x+1}}{x^2 - 7x - 8} =$
   (e) $\lim_{x \to -3} \dfrac{x - 5}{x + 3} =$
   (f) $\lim_{x \to 8} -\dfrac{1}{(x-8)^2} =$

3. Suppose that $f(x) = \dfrac{x+7}{x-3}$. Compute the difference quotient

$$\frac{f(x+h) - f(x)}{h}.$$

Simplify your answer until the h in the denominator cancels.

4. Sketch the graph of $f(x) = 1 + 4\sqrt{2x-2}$. Mark several points on your graph.

5. Consider the two functions

$$f(x) = \frac{1+x}{1-x} \qquad \text{and} \qquad g(x) = \frac{1}{x}.$$

(a) Compute $f \circ g(x)$. Simplify your answer to a single fraction. State the Domain.

(b) Compute $g \circ f(x)$. Simplify your answer to a single fraction. State the Domain.

(c) Compute $f \circ f(x)$. Simplify your answer to a single fraction. State the Domain.

6. Consider the function defined by

$$f(x) = \begin{cases} 2, & x \geq 11, \\ \sqrt{x-7}, & 7 < x < 11, \\ 1, & x = 7, \\ 7 - x, & 0 < x < 7, \\ 16 - x^2, & -4 < x \leq 0, \\ \dfrac{1}{x+4}, & x < -4. \end{cases}$$

(a) Carefully sketch the graph of $f(x)$.

(b) State the Domain of the function $f(x)$.

(c) Compute $\lim\limits_{x \to -4} f(x) =$

(d) Compute $\lim\limits_{x \to 0} f(x) =$

(e) Compute $\lim\limits_{x \to 7} f(x) =$

(f) Compute $\lim\limits_{x \to 11} f(x) =$

(g) State the value(s) at which f is discontinuous. Justify your answer(s) using the *definition of continuity* discussed in class.

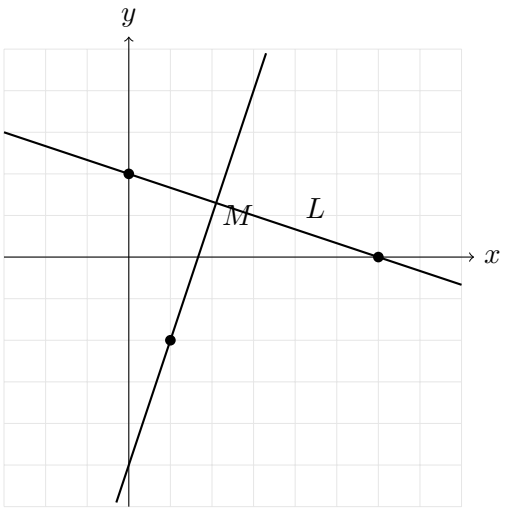
Math 105: Midterm 1 Practice Exam Solutions

1. Consider the line L given by $x + 3y = 6$.

   (a)
   $$x + 3y = 6 \quad \Longrightarrow \quad y = -\frac{1}{3}x + 2.$$
   Thus L has slope $-\frac{1}{3}$, y-intercept $(0, 2)$, and x-intercept $(6, 0)$.

   (b) A perpendicular line has slope 3. Through $(1, -2)$,
   $$y + 2 = 3(x - 1),$$
   so
   $$\boxed{y = 3x - 5}.$$

   (c) One possible sketch is:



2. Evaluate each limit.

   (a)
   $$\lim_{x \to -7} \frac{x^2 + 5x - 14}{x^2 - 4x + 4}.$$
   Direct substitution is allowed because the denominator is nonzero:
   $$\frac{49 - 35 - 14}{49 + 28 + 4} = 0.$$
   Thus
   $$\boxed{0}.$$

   (b)
   $$\lim_{x \to 4} \frac{x^2 - 9x + 20}{|4 - x|} = \lim_{x \to 4} \frac{(x - 4)(x - 5)}{|x - 4|}.$$
   As $x \to 4^+$, the limit is -1; as $x \to 4^-$, the limit is 1. Therefore
   $$\boxed{\text{DNE}}.$$

(c)

$$\lim_{x\to 2} \frac{x^2+5x-14}{x^2-4x+4} = \lim_{x\to 2} \frac{(x+7)(x-2)}{(x-2)^2} = \lim_{x\to 2} \frac{x+7}{x-2}.$$

Hence

$$\lim_{x\to 2^-} \frac{x+7}{x-2} = -\infty, \qquad \lim_{x\to 2^+} \frac{x+7}{x-2} = +\infty.$$

Thus

$$\boxed{\text{DNE}}.$$

(d)

$$\lim_{x\to 8} \frac{3-\sqrt{x+1}}{x^2-7x-8}.$$

Rationalize:

$$\frac{3-\sqrt{x+1}}{(x-8)(x+1)} \cdot \frac{3+\sqrt{x+1}}{3+\sqrt{x+1}} = -\frac{1}{(x+1)(3+\sqrt{x+1})}.$$

Therefore

$$\boxed{-\frac{1}{54}}.$$

(e)

$$\lim_{x\to -3} \frac{x-5}{x+3}.$$

Since

$$\lim_{x\to -3^-} \frac{x-5}{x+3} = +\infty, \qquad \lim_{x\to -3^+} \frac{x-5}{x+3} = -\infty,$$

we get

$$\boxed{\text{DNE}}.$$

(f)

$$\lim_{x\to 8} -\frac{1}{(x-8)^2}.$$

Because $(x-8)^2 \to 0^+$,

$$\boxed{-\infty}.$$

3. Suppose

$$f(x) = \frac{x+7}{x-3}.$$

Then

$$\frac{f(x+h)-f(x)}{h} = \frac{(x+h+7)(x-3)-(x+7)(x+h-3)}{h(x+h-3)(x-3)}.$$

The numerator simplifies to $-10h$, so

$$\boxed{\frac{f(x+h)-f(x)}{h} = -\frac{10}{(x+h-3)(x-3)}}.$$

4. Sketch

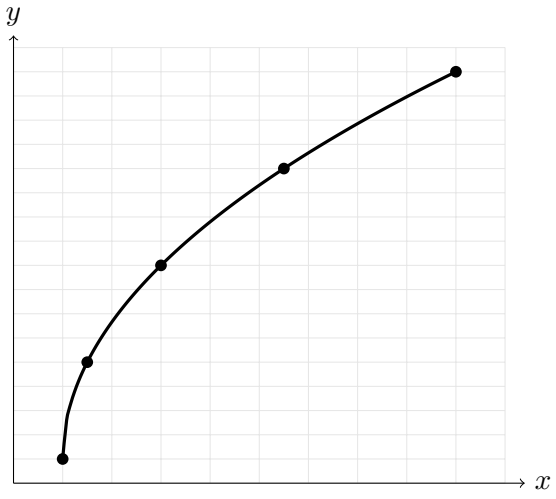
$$f(x) = 1 + 4\sqrt{2x - 2}.$$

The square root requires $x \geq 1$, and the graph begins at

$$\boxed{(1, 1)}.$$

Some useful points are

x	1	$\frac{3}{2}$	3	$\frac{11}{2}$	9
$f(x)$	1	5	9	13	17



5. Let

$$f(x) = \frac{1 + x}{1 - x}, \qquad g(x) = \frac{1}{x}.$$

(a)

$$(f \circ g)(x) = \frac{1 + \frac{1}{x}}{1 - \frac{1}{x}} = \boxed{\frac{x + 1}{x - 1}}.$$

The domain is

$$\boxed{\mathbb{R} \setminus \{0, 1\}}.$$

(b)

$$(g \circ f)(x) = \frac{1}{f(x)} = \boxed{\frac{1 - x}{1 + x}}.$$

The domain is

$$\boxed{\mathbb{R} \setminus \{-1, 1\}}.$$

(c)

$$(f \circ f)(x) = \frac{1 + \frac{1+x}{1-x}}{1 - \frac{1+x}{1-x}} = \boxed{-\frac{1}{x}}.$$

The domain of the original composition is

$$\boxed{\mathbb{R} \setminus \{0, 1\}}.$$

6. Consider

$$f(x) = \begin{cases} 2, & x \geq 11, \\ \sqrt{x-7}, & 7 < x < 11, \\ 1, & x = 7, \\ 7 - x, & 0 < x < 7, \\ 16 - x^2, & -4 < x \leq 0, \\ \dfrac{1}{x+4}, & x < -4. \end{cases}$$

(a) The sketch consists of:

- $y = \frac{1}{x+4}$ for $x < -4$, with vertical asymptote $x = -4$;
- $y = 16 - x^2$ for $-4 < x \leq 0$, open at $(-4, 0)$, closed at $(0, 16)$;
- $y = 7 - x$ for $0 < x < 7$, open at both endpoints;
- the filled point $(7, 1)$;
- $y = \sqrt{x-7}$ for $7 < x < 11$;
- $y = 2$ for $x \geq 11$, including $(11, 2)$.

(b)

$$\boxed{\text{Domain } = (-\infty, -4) \cup (-4, \infty)}.$$

(c)

$$\lim_{x \to -4^-} f(x) = -\infty, \qquad \lim_{x \to -4^+} f(x) = 0,$$

so

$$\boxed{\lim_{x \to -4} f(x) = \text{DNE}}.$$

(d)

$$\lim_{x \to 0^-} f(x) = 16, \qquad \lim_{x \to 0^+} f(x) = 7,$$

so

$$\boxed{\lim_{x \to 0} f(x) = \text{DNE}}.$$

(e)

$$\lim_{x \to 7^-} f(x) = 0, \qquad \lim_{x \to 7^+} f(x) = 0,$$

so

$$\boxed{\lim_{x \to 7} f(x) = 0}.$$

However, $f(7) = 1$.

(f)

$$\lim_{x \to 11^-} f(x) = 2, \qquad \lim_{x \to 11^+} f(x) = 2,$$

so

$$\boxed{\lim_{x \to 11} f(x) = 2}.$$

Also $f(11) = 2$.

(g) The discontinuities are

$$\boxed{x = -4,\ 0,\ 7}.$$

At $x = -4$, $f(-4)$ is undefined and the two-sided limit does not exist.
At $x = 0$, $f(0) = 16$, but the two-sided limit does not exist.
At $x = 7$, the limit exists and equals 0, but $f(7) = 1$, so

$$\lim_{x \to 7} f(x) \neq f(7).$$

Math 105, Fall 2026, Worksheet 6

Recall the **limit definition of the derivative.**

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}}$$

The value $f'(a)$ is the slope of the tangent line at the point $(a, f(a))$.

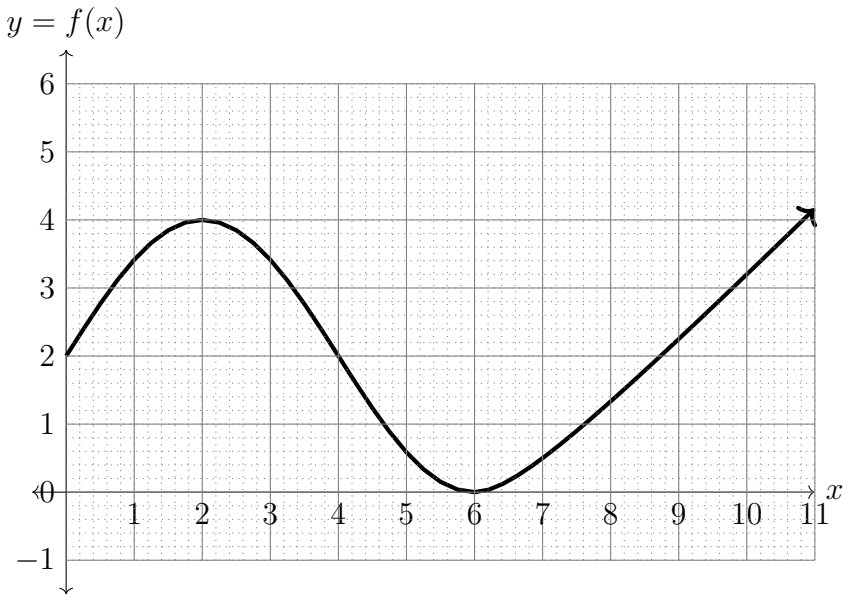
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1. Suppose that $f(x) = x^2$.

   (a) Compute $f'(1)$ using the limit definition. (Set $a = 1$ in the definition)

   (b) Write the **equation of the tangent line** to the curve $y = f(x)$ at the point where $x = 1$.

2. Consider the function $f(x)$ whose graph is



   (a) For which values of x is $f'(x)$ equal to 0? Where is it positive? Where is it negative? (For positive and negative, express your answer in interval notation.)

   (b) Sketch a little piece of the tangent line to this curves at $x = 0, 2, 4, 6, 8$, and 10. Using these sketches, approximate the value of $f'(x)$ at each of these points (no need to be too exact, just get a rough estimate.)

3. Suppose that $f(x) = 5 - 6x + 4x^2$.

   (a) Compute $f'(1)$ using the limit definition. (Set $a = 1$ in the definition)

   (b) Write the **equation of the tangent line** to the curve $y = f(x)$ at the point where $x = 1$.

   We can also think of the derivative as a function:

$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

   When taking the limit, we need to be careful distinguishing between x and h; the limit is with respect to h, so (after doing some manipulations) when we do direct substitution we should set $h = 0$.

4. For each of the following, find $f'(x)$ using the *limit definition of the derivative.*

   (a) $f(x) = x^3$

   (b) $f(x) = \sqrt{x}$

   (c) $f(x) = \dfrac{1}{x}$

   (d) $f(x) = \dfrac{x+1}{x-1}$

   (e) $f(x) = \dfrac{1}{\sqrt{x}}$

5. When simplifying a sum or difference of two complicated expressions, it often saves a lot of work to look for common factors before proceeding. This problem gives some practice identifying common factors.

   (a) Consider the expression

$$9x^2y + 2xy^3.$$

   Simplify this expression by factoring out xy.

   (b) Consider the expression

$$2(x+1)^2y^3 - 8(x+1)y^5.$$

   What is the biggest common factor that can be factored out? Factor this out and simplify the expression.

   (c) Simplify the expression

$$3(x+1)^2(1-2x)^4 + (x+1)^3 4(1-2x)^3(-2).$$

   This problem arises in a derivative computation that we will do in a couple weeks.

Math 105, Fall 2026, Worksheet 6 Solutions

1. Suppose that $f(x) = x^2$.

   (a) Using the limit definition at $a = 1$,

$$\begin{aligned} f'(1) &= \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h\to 0} \frac{(1+h)^2 - 1}{h} \\ &= \lim_{h\to 0} \frac{2h + h^2}{h} \\ &= \lim_{h\to 0} (2 + h) = \boxed{2}. \end{aligned}$$

   (b) The tangent line passes through $(1, f(1)) = (1, 1)$ and has slope $f'(1) = 2$. Thus

$$y - 1 = 2(x - 1),$$

   so

$$\boxed{y = 2x - 1}.$$

2. Consider the function shown in the worksheet.

   (a) The derivative is zero where the graph has a horizontal tangent. From the graph,

$$\boxed{f'(x) = 0 \text{ at } x = 2 \text{ and } x = 6.}$$

   The graph is increasing on $(0, 2)$ and again on $(6, 11)$, so on the portion shown,

$$\boxed{f'(x) > 0 \text{ on } (0, 2) \cup (6, 11).}$$

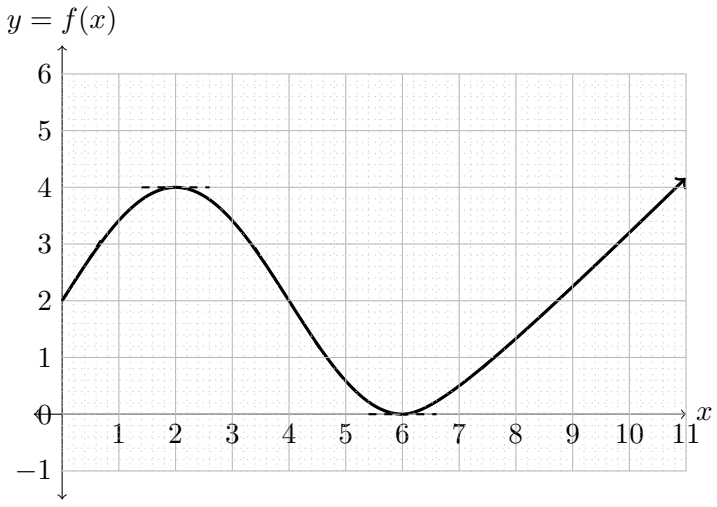
   The graph is decreasing on $(2, 6)$, so

$$\boxed{f'(x) < 0 \text{ on } (2, 6).}$$

   (b) One possible set of tangent-line sketches is shown below. Reading off their slopes gives roughly

x	0	2	4	6	8	10
$f'(x)$	1.6	0	-1.6	0	0.9	1.0

   The point here is the sign and rough size of the slope; small differences in the estimates are fine.



3. Suppose that $f(x) = 5 - 6x + 4x^2$.

(a) First, $f(1) = 3$. Also,

$$\begin{aligned} f(1+h) &= 5 - 6(1+h) + 4(1+h)^2 \\ &= 3 + 2h + 4h^2. \end{aligned}$$

Therefore

$$\begin{aligned} f'(1) &= \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h\to 0} \frac{2h + 4h^2}{h} \\ &= \lim_{h\to 0} (2 + 4h) = \boxed{2}. \end{aligned}$$

(b) The tangent line passes through $(1, 3)$ and has slope 2. Hence

$$y - 3 = 2(x - 1),$$

so

$$\boxed{y = 2x + 1}.$$

4. Find each derivative using the limit definition.

(a) If $f(x) = x^3$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\ &= \lim_{h\to 0} (3x^2 + 3xh + h^2) \\ &= \boxed{3x^2}. \end{aligned}$$

(b) If $f(x) = \sqrt{x}$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \boxed{\frac{1}{2\sqrt{x}}}, \qquad x > 0. \end{aligned}$$

(c) If $f(x) = \dfrac{1}{x}$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\ &= \lim_{h\to 0} \frac{-h}{hx(x+h)} \\ &= \lim_{h\to 0} -\frac{1}{x(x+h)} \\ &= \boxed{-\frac{1}{x^2}}, \qquad x \neq 0. \end{aligned}$$

(d) If $f(x) = \dfrac{x+1}{x-1}$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\frac{x+h+1}{x+h-1} - \frac{x+1}{x-1}}{h} \\ &= \lim_{h\to 0} \frac{-2h}{h(x+h-1)(x-1)} \\ &= \lim_{h\to 0} -\frac{2}{(x+h-1)(x-1)} \\ &= \boxed{-\frac{2}{(x-1)^2}}, \qquad x \neq 1. \end{aligned}$$

(e) If $f(x) = \dfrac{1}{\sqrt{x}}$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h} \\ &= \lim_{h\to 0} \frac{\sqrt{x} - \sqrt{x+h}}{h\sqrt{x}\sqrt{x+h}}. \end{aligned}$$

Rationalizing the numerator gives

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{-1}{\sqrt{x}\sqrt{x+h}\left(\sqrt{x} + \sqrt{x+h}\right)} \\ &= \boxed{-\frac{1}{2x^{3/2}}}, \qquad x > 0. \end{aligned}$$

5. Practice factoring common factors.

(a)

$$9x^2y + 2xy^3 = \boxed{xy(9x + 2y^2)}.$$

(b) The largest common factor is $2(x+1)y^3$. Thus

$$2(x+1)^2y^3 - 8(x+1)y^5 = \boxed{2(x+1)y^3\big((x+1) - 4y^2\big)}.$$

(c) The common factor is $(x+1)^2(1-2x)^3$:

$$\begin{aligned} &3(x+1)^2(1-2x)^4 + (x+1)^3 4(1-2x)^3(-2) \\ &\quad = (x+1)^2(1-2x)^3\big(3(1-2x) - 8(x+1)\big) \\ &\quad = (x+1)^2(1-2x)^3(-14x-5). \end{aligned}$$

Therefore

$$\boxed{-(14x+5)(x+1)^2(1-2x)^3}.$$

Math 105, Fall 2026, Worksheet 7

Differentiate the following functions.

1. $f(x) = 5x^{20} + 2x^{10} + 25$.

2. $g(t) = \dfrac{6}{\sqrt[3]{t^5}}$.

3. $F(y) = \dfrac{4y^2 + 5}{2 - 3y}$.

4. $y = \dfrac{\sqrt{x} - 6}{\sqrt{x} + 6}$.

5. Differentiate $h(x) = \big(\sqrt{x} - 6\big)\big(\sqrt{x} + 6\big)$ using the product rule.

6. Differentiate $h(x) = \big(\sqrt{x} - 6\big)\big(\sqrt{x} + 6\big)$ by multiplying out **before** differentiating.

7. Differentiate the following function. $\left(x^2 - \dfrac{5}{x^2}\right)\left(3x + \sqrt{x} + \dfrac{1}{x}\right)$.

8. Differentiate $\dfrac{x^2 G(x)}{3x - 1}$, where $G(x)$ is an unspecified differentiable function.
   Your answer should involve both $G(x)$ and $G'(x)$.

9. Let $f(x)$ and $g(x)$ be differentiable functions with the following table of values:

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	4	-3	2	7
2	-2	6	1	5
3	3	-2	-1	0

   Let

$$h(x) = f(x)g(x), \qquad k(x) = \frac{g(x)}{f(x)}, \qquad P(x) = f(x)f(x).$$

   Compute

$$h'(1), \qquad k'(3), \qquad P'(2).$$

10. Find the equation of the tangent line to the curve

$$y = \frac{x^2 - 4x + 3}{x^3 + 2x - 4}$$

at the point where $x = 1$.

11. Compute the derivative of

$$f(x) = \frac{x}{x-1} + \frac{x}{x+1}.$$

After computing the derivative, simplify your answer to a single fraction.

12. Compute the derivatives of the following functions. Simplify when possible.

(a) $y = \left(\sqrt{x} + \dfrac{1}{x}\right)^9$.

(b) $f(x) = \dfrac{1}{\sqrt{x^2 - 5x + 3}}$.

(c) $g(x) = \left(\dfrac{1}{x^3} + 7x\right)^4 \left(x^4 - \dfrac{1}{x^7}\right)^{-5}$.

(d) $h(x) = \sqrt{\dfrac{x+5}{5-x}}$.

(e) $F(x) = (7x - 3)^4(5x + 2)^6$.

Math 105, Fall 2026, Worksheet 7 Solutions

1.

$$f(x) = 5x^{20} + 2x^{10} + 25.$$

Using the power rule,

$$\boxed{f'(x) = 100x^{19} + 20x^9}.$$

2.

$$g(t) = \frac{6}{\sqrt[3]{t^5}} = 6t^{-5/3}.$$

Therefore

$$g'(t) = 6\left(-\frac{5}{3}\right)t^{-8/3},$$

so

$$\boxed{g'(t) = -10t^{-8/3} = -\frac{10}{t^{8/3}}}.$$

3.

$$F(y) = \frac{4y^2 + 5}{2 - 3y}.$$

Using the quotient rule,

$$F'(y) = \frac{(8y)(2 - 3y) - (4y^2 + 5)(-3)}{(2 - 3y)^2}.$$

Thus

$$\boxed{F'(y) = \frac{-12y^2 + 16y + 15}{(2 - 3y)^2}}.$$

4.

$$y = \frac{\sqrt{x} - 6}{\sqrt{x} + 6}.$$

Using the quotient rule,

$$y' = \frac{\frac{1}{2\sqrt{x}}(\sqrt{x} + 6) - (\sqrt{x} - 6)\frac{1}{2\sqrt{x}}}{(\sqrt{x} + 6)^2}.$$

The numerator simplifies to

$$\frac{12}{2\sqrt{x}} = \frac{6}{\sqrt{x}},$$

so

$$\boxed{y' = \frac{6}{\sqrt{x}(\sqrt{x} + 6)^2}}.$$

5. Differentiate

$$h(x) = \big(\sqrt{x} - 6\big)\big(\sqrt{x} + 6\big)$$

using the product rule.

We have

$$\begin{aligned} h'(x) &= \frac{1}{2\sqrt{x}}(\sqrt{x} + 6) + (\sqrt{x} - 6)\frac{1}{2\sqrt{x}} \\ &= \frac{(\sqrt{x} + 6) + (\sqrt{x} - 6)}{2\sqrt{x}} \\ &= \frac{2\sqrt{x}}{2\sqrt{x}}. \end{aligned}$$

Therefore

$$\boxed{h'(x) = 1} \qquad (x > 0).$$

6. Differentiate

$$h(x) = \big(\sqrt{x} - 6\big)\big(\sqrt{x} + 6\big)$$

by multiplying out first.

Using the difference of squares,

$$h(x) = x - 36.$$

Therefore

$$\boxed{h'(x) = 1} \qquad (x > 0).$$

7. Let

$$f(x) = \left(x^2 - \frac{5}{x^2}\right)\left(3x + \sqrt{x} + \frac{1}{x}\right).$$

Write the factors using exponents:

$$x^2 - 5x^{-2}, \qquad 3x + x^{1/2} + x^{-1}.$$

Their derivatives are

$$2x + 10x^{-3} \qquad \text{and} \qquad 3 + \frac{1}{2\sqrt{x}} - \frac{1}{x^2}.$$

By the product rule,

$$\boxed{\begin{aligned} f'(x) = &\left(2x + \frac{10}{x^3}\right)\left(3x + \sqrt{x} + \frac{1}{x}\right) \\ &+ \left(x^2 - \frac{5}{x^2}\right)\left(3 + \frac{1}{2\sqrt{x}} - \frac{1}{x^2}\right). \end{aligned}}$$

8. Let

$$F(x) = \frac{x^2 G(x)}{3x - 1}.$$

The derivative of the numerator is

$$\frac{d}{dx}\big(x^2 G(x)\big) = 2xG(x) + x^2 G'(x).$$

Using the quotient rule,

$$F'(x) = \frac{\big(2xG(x) + x^2 G'(x)\big)(3x - 1) - 3x^2 G(x)}{(3x - 1)^2}.$$

Thus

$$\boxed{F'(x) = \frac{x^2(3x - 1)G'(x) + (3x^2 - 2x)G(x)}{(3x - 1)^2}}.$$

9. Recall

$$h(x) = f(x)g(x), \qquad k(x) = \frac{g(x)}{f(x)}, \qquad P(x) = f(x)f(x).$$

For h,

$$h'(x) = f'(x)g(x) + f(x)g'(x).$$

At $x = 1$,

$$h'(1) = (-3)(2) + (4)(7) = -6 + 28,$$

so

$$\boxed{h'(1) = 22}.$$

For k,

$$k'(x) = \frac{g'(x)f(x) - g(x)f'(x)}{f(x)^2}.$$

At $x = 3$,

$$k'(3) = \frac{(0)(3) - (-1)(-2)}{3^2} = -\frac{2}{9},$$

so

$$\boxed{k'(3) = -\frac{2}{9}}.$$

Finally,

$$P'(x) = f'(x)f(x) + f(x)f'(x) = 2f(x)f'(x).$$

At $x = 2$,

$$P'(2) = 2(-2)(6),$$

so

$$\boxed{P'(2) = -24}.$$

10. Let

$$y = \frac{x^2 - 4x + 3}{x^3 + 2x - 4}.$$

At $x = 1$,

$$y(1) = \frac{1 - 4 + 3}{1 + 2 - 4} = 0,$$

so the point is $(1, 0)$.

Using the quotient rule,

$$y' = \frac{(2x - 4)(x^3 + 2x - 4) - (x^2 - 4x + 3)(3x^2 + 2)}{(x^3 + 2x - 4)^2}.$$

At $x = 1$,

$$y'(1) = \frac{(-2)(-1) - (0)(5)}{(-1)^2} = 2.$$

Thus the tangent line has slope 2 and passes through $(1, 0)$:

$$y - 0 = 2(x - 1).$$

Therefore

$$\boxed{y = 2x - 2}.$$

11.

$$f(x) = \frac{x}{x - 1} + \frac{x}{x + 1}.$$

Differentiate each fraction:

$$f'(x) = -\frac{1}{(x - 1)^2} + \frac{1}{(x + 1)^2}.$$

Combining these into one fraction,

$$\begin{aligned} f'(x) &= \frac{(x - 1)^2 - (x + 1)^2}{(x - 1)^2(x + 1)^2} \\ &= \frac{-4x}{(x^2 - 1)^2}. \end{aligned}$$

Thus

$$\boxed{f'(x) = -\frac{4x}{(x^2 - 1)^2}}.$$

12. Compute the derivatives.

(a)

$$y = \left(\sqrt{x} + \frac{1}{x}\right)^9.$$

By the chain rule,

$$\boxed{y' = 9\left(\sqrt{x} + \frac{1}{x}\right)^8 \left(\frac{1}{2\sqrt{x}} - \frac{1}{x^2}\right)}.$$

(b)

$$f(x) = \frac{1}{\sqrt{x^2 - 5x + 3}} = (x^2 - 5x + 3)^{-1/2}.$$

Using the chain rule,

$$f'(x) = -\frac{1}{2}(x^2 - 5x + 3)^{-3/2}(2x - 5).$$

Therefore

$$\boxed{f'(x) = \frac{5 - 2x}{2(x^2 - 5x + 3)^{3/2}}}.$$

(c)

$$g(x) = \left(\frac{1}{x^3} + 7x\right)^4 \left(x^4 - \frac{1}{x^7}\right)^{-5}.$$

Let

$$A(x) = \left(x^{-3} + 7x\right)^4, \qquad B(x) = \left(x^4 - x^{-7}\right)^{-5}.$$

Then

$$A'(x) = 4(x^{-3} + 7x)^3(-3x^{-4} + 7),$$

and

$$B'(x) = -5(x^4 - x^{-7})^{-6}(4x^3 + 7x^{-8}).$$

Using the product rule,

$$\boxed{\begin{aligned} g'(x) = {} & 4(x^{-3} + 7x)^3(-3x^{-4} + 7)(x^4 - x^{-7})^{-5} \\ & - 5(x^{-3} + 7x)^4(x^4 - x^{-7})^{-6}(4x^3 + 7x^{-8}). \end{aligned}}$$

(d)

$$h(x) = \sqrt{\frac{x+5}{5-x}} = \left(\frac{x+5}{5-x}\right)^{1/2}.$$

First,

$$\frac{d}{dx}\left(\frac{x+5}{5-x}\right) = \frac{(5-x) + (x+5)}{(5-x)^2} = \frac{10}{(5-x)^2}.$$

Thus

$$h'(x) = \frac{1}{2}\left(\frac{x+5}{5-x}\right)^{-1/2} \frac{10}{(5-x)^2}.$$

So

$$\boxed{h'(x) = \frac{5}{(5-x)^2}\sqrt{\frac{5-x}{x+5}}}.$$

(e)

$$F(x) = (7x - 3)^4(5x + 2)^6.$$

Using the product and chain rules,

$$\begin{aligned} F'(x) &= 28(7x-3)^3(5x+2)^6 \\ &\quad + 30(7x-3)^4(5x+2)^5. \end{aligned}$$

Factoring,

$$\boxed{F'(x) = (7x-3)^3(5x+2)^5(350x-34)}.$$

Instructor: David Zureick-Brown
Fall 2026

Amherst College
Dept. of Mathematics

Math 105 ∼ Calculus with Algebra: Midterm 2

1. Compute the derivative of
$$f(x) = \frac{3x-1}{2-5x}$$
two different ways:
   - First use the **limit definition of the derivative**.
   - Second use the Chain Rule.

   Next simplify your answer from the first part.

2. Find the equation of the tangent line to this curve
$$y = \left(6x + \sqrt{8+x^2}\right)^{2/3}$$
at the point where $x = 1$.

3. Consider
$$f(x) = \frac{5x}{1+x}.$$
Compute $f'(0)$, $f'(1)$, and $f'(2)$.

4. Find **all** x-coordinates at which the graph of the following function has horizontal tangent lines. I recommend simplifying your derivative first. (In this case, that will mean factoring out any common terms.)
$$f(x) = (5+3x^2)^8(7-x^2)^3.$$

5. **Implicit Differentiation Practice**

   (a) Consider the equation
$$\frac{x}{y+1} = x^2 - y^2.$$
   Compute the equation of the tangent line at the point $(1, 0)$.

   (b) Find the equation of the tangent line to the curve
$$4(x+y)^2 = x^2y^2$$
   at the point $(-2, 1)$.

6. Compute the derivative of each of the following functions. For these problems, you do **NOT** need to simplify your derivative.

(a)
$$y = \frac{5}{6}x + x^{5/6} + \frac{1}{x^{5/6}} + \sqrt{5x+6} + \frac{1}{\sqrt{5x+6}}.$$

(b)
$$y = \left(\frac{2\sqrt{x} + x^3}{x^{2/3} + \frac{2}{3}x}\right)^{2/3}.$$

(c)
$$f(x) = \left(\frac{3}{x^2} - \frac{2}{x^3}\right)^9 \sqrt{5 - x^2}.$$

(d)
$$y = \frac{\frac{1}{x} - 6x^3}{\sqrt{7x + x^8}}.$$

Math 105: Calculus with Algebra – Fall 2026

Midterm 2 Practice Exam Solutions

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1. Compute the derivative of

$$f(x) = \frac{3x-1}{2-5x}$$

two different ways.

Using the limit definition:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \frac{1}{h}\left(\frac{3x+3h-1}{2-5x-5h} - \frac{3x-1}{2-5x}\right) \\ &= \lim_{h\to 0} \frac{(3x+3h-1)(2-5x)-(3x-1)(2-5x-5h)}{h(2-5x-5h)(2-5x)}. \end{aligned}$$

The numerator simplifies to

$$3h(2-5x)+5h(3x-1) = h,$$

so

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{1}{(2-5x-5h)(2-5x)} \\ &= \boxed{\frac{1}{(2-5x)^2}}. \end{aligned}$$

Using differentiation rules and the Chain Rule: Write

$$f(x) = (3x-1)(2-5x)^{-1}.$$

Then

$$\begin{aligned} f'(x) &= 3(2-5x)^{-1} + (3x-1)(-1)(2-5x)^{-2}(-5) \\ &= \frac{3(2-5x)+5(3x-1)}{(2-5x)^2} \\ &= \boxed{\frac{1}{(2-5x)^2}}. \end{aligned}$$

2. Find the equation of the tangent line to

$$y = \left(6x + \sqrt{8+x^2}\right)^{2/3}$$

at the point where $x = 1$.

First find the point:

$$y(1) = \left(6+\sqrt{9}\right)^{2/3} = 9^{2/3}.$$

Using the Chain Rule,

$$y' = \frac{2}{3}\left(6x+\sqrt{8+x^2}\right)^{-1/3}\left(6+\frac{x}{\sqrt{8+x^2}}\right).$$

Therefore

$$y'(1) = \frac{2}{3}\,9^{-1/3}\left(6 + \frac{1}{3}\right)$$
$$= \frac{38}{9\sqrt[3]{9}}.$$

Thus the tangent line is

$$\boxed{y - 9^{2/3} = \frac{38}{9\sqrt[3]{9}}(x-1)}.$$

3. Consider

$$f(x) = \frac{5x}{1+x}.$$

By the quotient rule,

$$f'(x) = \frac{5(1+x) - 5x}{(1+x)^2} = \frac{5}{(1+x)^2}.$$

Hence

$$\boxed{f'(0) = 5}, \qquad \boxed{f'(1) = \frac{5}{4}}, \qquad \boxed{f'(2) = \frac{5}{9}}.$$

4. Find all x-coordinates where

$$f(x) = (5 + 3x^2)^8 (7 - x^2)^3$$

has a horizontal tangent.

Using the product and chain rules,

$$f'(x) = 8(5+3x^2)^7(6x)(7-x^2)^3 + (5+3x^2)^8\,3(7-x^2)^2(-2x).$$

Factor:

$$f'(x) = 6x(5+3x^2)^7(7-x^2)^2\left(8(7-x^2) - (5+3x^2)\right)$$
$$= 6x(5+3x^2)^7(7-x^2)^2(51 - 11x^2).$$

Since $5 + 3x^2$ is never zero, $f'(x) = 0$ when

$$x = 0, \qquad x = \pm\sqrt{7}, \qquad x = \pm\sqrt{\frac{51}{11}}.$$

Thus the x-coordinates are

$$\boxed{0,\ \pm\sqrt{7},\ \pm\sqrt{\frac{51}{11}}}.$$

5. **Implicit Differentiation Practice**

(a) Consider

$$\frac{x}{y+1} = x^2 - y^2.$$

Differentiate implicitly:

$$\frac{(y+1) - xy'}{(y+1)^2} = 2x - 2yy'.$$

At $(1,0)$ this becomes

$$1 - y' = 2,$$

so

$$y' = -1.$$

The tangent line through $(1,0)$ is therefore

$$y = -1(x-1),$$

so

$$\boxed{y = -x + 1}.$$

(b) Consider

$$4(x+y)^2 = x^2y^2.$$

Differentiate implicitly:

$$8(x+y)(1+y') = 2xy^2 + 2x^2yy'.$$

At $(-2,1)$,

$$-8(1+y') = -4 + 8y'.$$

Thus

$$-8 - 8y' = -4 + 8y', \qquad -4 = 16y', \qquad y' = -\frac{1}{4}.$$

The tangent line is

$$y - 1 = -\frac{1}{4}(x+2),$$

so

$$\boxed{y - 1 = -\frac{1}{4}(x+2)}.$$

6. Compute the derivative of each function. No simplification is required.

(a)

$$y = \frac{5}{6}x + x^{5/6} + \frac{1}{x^{5/6}} + \sqrt{5x+6} + \frac{1}{\sqrt{5x+6}}.$$

Rewrite powers where helpful:

$$y = \frac{5}{6}x + x^{5/6} + x^{-5/6} + (5x+6)^{1/2} + (5x+6)^{-1/2}.$$

Therefore

$$\boxed{y' = \frac{5}{6} + \frac{5}{6}x^{-1/6} - \frac{5}{6}x^{-11/6} + \frac{5}{2}(5x+6)^{-1/2} - \frac{5}{2}(5x+6)^{-3/2}}.$$

(b)

$$y = \left(\frac{2\sqrt{x} + x^3}{x^{2/3} + \frac{2}{3}x}\right)^{2/3}.$$

Let

$$u(x) = 2\sqrt{x} + x^3, \qquad v(x) = x^{2/3} + \frac{2}{3}x.$$

Then

$$u'(x) = \frac{1}{\sqrt{x}} + 3x^2, \qquad v'(x) = \frac{2}{3}x^{-1/3} + \frac{2}{3}.$$

By the Chain Rule and Quotient Rule,

$$\boxed{y' = \frac{2}{3}\left(\frac{u}{v}\right)^{-1/3} \frac{u'v - uv'}{v^2}},$$

where u, u', v, v' are as above. Equivalently,

$$\boxed{y' = \frac{2}{3}\left(\frac{2\sqrt{x} + x^3}{x^{2/3} + \frac{2}{3}x}\right)^{-1/3} \cdot \frac{\left(\frac{1}{\sqrt{x}} + 3x^2\right)\left(x^{2/3} + \frac{2}{3}x\right) - \left(2\sqrt{x} + x^3\right)\left(\frac{2}{3}x^{-1/3} + \frac{2}{3}\right)}{\left(x^{2/3} + \frac{2}{3}x\right)^2}}.$$

(c)

$$f(x) = \left(\frac{3}{x^2} - \frac{2}{x^3}\right)^9 \sqrt{5 - x^2}.$$

Using the product and chain rules,

$$\boxed{f'(x) = 9\left(\frac{3}{x^2} - \frac{2}{x^3}\right)^8 \left(-\frac{6}{x^3} + \frac{6}{x^4}\right)\sqrt{5 - x^2} + \left(\frac{3}{x^2} - \frac{2}{x^3}\right)^9 \left(-\frac{x}{\sqrt{5 - x^2}}\right).}$$

(d)

$$y = \frac{\frac{1}{x} - 6x^3}{\sqrt{7x + x^8}}.$$

Let

$$u(x) = \frac{1}{x} - 6x^3, \qquad v(x) = \sqrt{7x + x^8}.$$

Then

$$u'(x) = -\frac{1}{x^2} - 18x^2,$$

and

$$v'(x) = \frac{7 + 8x^7}{2\sqrt{7x + x^8}}.$$

By the quotient rule,

$$\boxed{y' = \frac{\left(-\frac{1}{x^2} - 18x^2\right)\sqrt{7x + x^8} - \left(\frac{1}{x} - 6x^3\right)\left(\frac{7 + 8x^7}{2\sqrt{7x + x^8}}\right)}{7x + x^8}}.$$

Math 105, Fall 2026, Worksheet 9

For problems 1-5, x, y, and z are all functions of a variable t. Differentiate each equation implicitly with respect to t.

1. $x^2y + 2y^3 = 7$.

2. $xyz = 12$.

3. $x^2y + yz^2 = 5$.

4. $\sqrt{x} + \sqrt{y} = z$.

5. $\frac{x}{y} + z^2 = 4$.

6. If $x^2 + y^2 + z^2 = 9$, $\dfrac{dx}{dt} = 5$, and $\dfrac{dy}{dt} = 4$, find $\dfrac{dz}{dt}$ when $(x, y, z) = (2, 2, 1)$.

7. Suppose $xy + z^2 = 10$.
   If $x = z = 2$, $\dfrac{dx}{dt} = 1$ and $\dfrac{dy}{dt} = -2$, find $\dfrac{dz}{dt}$.

8. Suppose $x + y + z(x - y) = 0$
   If $\dfrac{dz}{dt} = 1$, and $z = \dfrac{dx}{dt} = \dfrac{dy}{dt} = 0$, find x and y.

9. A snowball remains spherical while it melts. At the moment when the radius is 2 inches, the radius is shrinking at a rate of 1 inch per hour. How fast is the volume of the snowball decreasing at that moment?

10. A kite is flying 300 feet above the ground, and it is being blown horizontally away from the person on the ground holding the string. Of course, the person on the ground is having to let out more string as the kite moves away. At the moment when there is 500 feet of string between the person and the kite, the kite is moving horizontally at 10 feet per second. How fast is the person having to let out the string at that moment?

11. A stone is dropped from a bridge and falls down to the river 576 feet below. The stone's position above the water at time t seconds after its release is given in feet by $s(t) = 576 - 16t^2$.

    - How long does it take for the stone to hit the water?

    - What is the stone's velocity at the moment of impact?

12. An oil spill occurs at sea. The oil gushes out from an offshore derrick and forms a circle whose area increases at a rate of 100 ft^2/sec. How fast is the radius of the spill increasing when the spill is 120 feet wide?

13. A ladder is ten feet long, leaning against a vertical wall. The bottom starts to slide horizontally away from the wall. At the moment when the top of the ladder is eight feet above the ground, it is sliding down at four inches per second. How fast is the bottom of the ladder moving at that moment?

14. A water tank is in the shape of an inverted cone (i.e., circular opening at the top, vertex at the bottom). The diameter at the top is 12 feet, and the tank is 14 feet deep. Water is leaking out the bottom of the tank. When the water is 4 feet deep, the water level is dropping at a rate of 2 feet per minute. How fast is the water leaking out of the tank at that time?

    (Hint: the volume of a cone of height h and radius r is $V = \pi r^2 h/3$)

Math 105, Fall 2026, Worksheet 9 — Solution Guide

For Problems 1–5, write $x' = \dfrac{dx}{dt}$, $y' = \dfrac{dy}{dt}$, and $z' = \dfrac{dz}{dt}$.

1. $x^2y + 2y^3 = 7$.

   Differentiate both sides with respect to t:

   $$2xx'y + x^2y' + 6y^2y' = 0.$$

   Equivalently,

   $$2xy\frac{dx}{dt} + x^2\frac{dy}{dt} + 6y^2\frac{dy}{dt} = 0.$$

2. $xyz = 12$.

   Using the product rule,

   $$x'yz + xy'z + xyz' = 0.$$

   That is,

   $$yz\frac{dx}{dt} + xz\frac{dy}{dt} + xy\frac{dz}{dt} = 0.$$

3. $x^2y + yz^2 = 5$.

   Differentiate both products:

   $$2xx'y + x^2y' + y'z^2 + 2yzz' = 0.$$

4. $\sqrt{x} + \sqrt{y} = z$.

   Differentiate:

   $$\frac{x'}{2\sqrt{x}} + \frac{y'}{2\sqrt{y}} = z'.$$

   Thus

   $$\frac{1}{2\sqrt{x}}\frac{dx}{dt} + \frac{1}{2\sqrt{y}}\frac{dy}{dt} = \frac{dz}{dt}.$$

5. $\dfrac{x}{y} + z^2 = 4$.

   Using the quotient rule,

   $$\frac{yx' - xy'}{y^2} + 2zz' = 0.$$

6. If $x^2 + y^2 + z^2 = 9$, $\dfrac{dx}{dt} = 5$, and $\dfrac{dy}{dt} = 4$, find $\dfrac{dz}{dt}$ when $(x, y, z) = (2, 2, 1)$.

   Differentiate:

   $$2x\frac{dx}{dt} + 2y\frac{dy}{dt} + 2z\frac{dz}{dt} = 0.$$

Substitute the given values:

$$2(2)(5) + 2(2)(4) + 2(1)\frac{dz}{dt} = 0.$$

Hence

$$20 + 16 + 2\frac{dz}{dt} = 0,$$

so

$$\boxed{\frac{dz}{dt} = -18}.$$

7. Suppose $xy + z^2 = 10$. If $x = z = 2$, $\dfrac{dx}{dt} = 1$, and $\dfrac{dy}{dt} = -2$, find $\dfrac{dz}{dt}$.

First find y. Since $x = z = 2$,

$$2y + 4 = 10,$$

so $y = 3$.

Differentiate the original equation:

$$y\frac{dx}{dt} + x\frac{dy}{dt} + 2z\frac{dz}{dt} = 0.$$

Substitute:

$$3(1) + 2(-2) + 4\frac{dz}{dt} = 0.$$

Therefore

$$-1 + 4\frac{dz}{dt} = 0,$$

and

$$\boxed{\frac{dz}{dt} = \frac{1}{4}}.$$

8. Suppose $x + y + z(x - y) = 0$. If $\dfrac{dz}{dt} = 1$, and $z = \dfrac{dx}{dt} = \dfrac{dy}{dt} = 0$, find x and y.

Since $z = 0$, the original equation gives

$$x + y = 0.$$

Differentiate:

$$x' + y' + z'(x - y) + z(x' - y') = 0.$$

Using $x' = y' = z = 0$ and $z' = 1$ gives

$$x - y = 0.$$

Together,

$$x + y = 0, \qquad x - y = 0,$$

so

$$\boxed{x = 0, \quad y = 0}.$$

9. A snowball remains spherical while it melts. At the moment when the radius is 2 inches, the radius is shrinking at a rate of 1 inch per hour. How fast is the volume decreasing?

   For a sphere,

$$V = \frac{4}{3}\pi r^3.$$

   Differentiate:

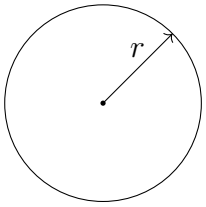
$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

   Here $r = 2$ and $\dfrac{dr}{dt} = -1$, so

$$\frac{dV}{dt} = 4\pi(2)^2(-1) = -16\pi.$$

   Thus the volume is decreasing at

$$\boxed{16\pi \text{ in}^3/\text{hr}}.$$



snowball

10. A kite is flying 300 feet above the ground. When there are 500 feet of string between the person and the kite, the kite is moving horizontally away at 10 ft/sec. How fast is the string being let out?

    Let x be the horizontal distance and s the length of the string. Then

$$x^2 + 300^2 = s^2.$$

    When $s = 500$,

$$x^2 = 500^2 - 300^2 = 160000,$$

    so $x = 400$.

    Differentiate:

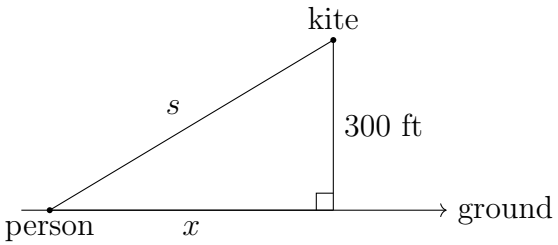
$$2x\frac{dx}{dt} = 2s\frac{ds}{dt}.$$

    Thus

$$400(10) = 500\frac{ds}{dt},$$

    so

$$\boxed{\frac{ds}{dt} = 8 \text{ ft/sec}}.$$



11. A stone is dropped from a bridge 576 feet above the water, with position

$$s(t) = 576 - 16t^2.$$

- The stone hits the water when $s(t) = 0$:

$$576 - 16t^2 = 0.$$

Thus $t^2 = 36$, and since $t \geq 0$,

$$\boxed{t = 6 \text{ sec}}.$$

- Velocity is the derivative of position:

$$v(t) = s'(t) = -32t.$$

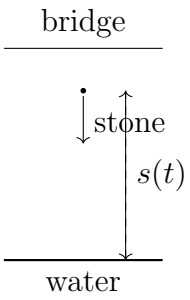
At impact,

$$v(6) = -32(6) = -192.$$

Therefore the velocity is

$$\boxed{-192 \text{ ft/sec}},$$

i.e. 192 ft/sec downward.



12. An oil spill forms a circle whose area increases at 100 ft^2/sec. How fast is its radius increasing when the spill is 120 feet wide?

Since the spill is 120 feet wide, its radius is $r = 60$ feet. The area is

$$A = \pi r^2.$$

Differentiate:

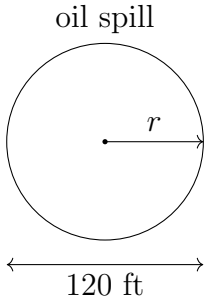
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}.$$

Substitute $\dfrac{dA}{dt} = 100$ and $r = 60$:

$$100 = 120\pi \frac{dr}{dt}.$$

Hence

$$\boxed{\frac{dr}{dt} = \frac{5}{6\pi} \text{ ft/sec}}.$$



13. A ten-foot ladder leans against a wall. When the top is 8 feet above the ground, it is sliding down at 4 inches/sec. How fast is the bottom moving?

Let x be the distance of the bottom from the wall and y the height of the top. Then

$$x^2 + y^2 = 100.$$

When $y = 8$,

$$x^2 + 64 = 100,$$

so $x = 6$ feet.

The top moves downward at 4 inches/sec, so

$$\frac{dy}{dt} = -\frac{1}{3} \text{ ft/sec}.$$

Differentiate:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0.$$

Therefore

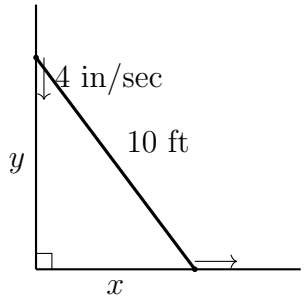
$$6\frac{dx}{dt} + 8\left(-\frac{1}{3}\right) = 0,$$

so

$$\frac{dx}{dt} = \frac{4}{9} \text{ ft/sec}.$$

Equivalently,

$$\boxed{\frac{dx}{dt} = \frac{16}{3} \text{ in/sec}}.$$



14. A water tank is an inverted cone with top diameter 12 feet and height 14 feet. When the water is 4 feet deep, its level is dropping at 2 ft/min. How fast is water leaking out?

Let h be the water depth and r the radius of the water surface. By similar triangles,

$$\frac{r}{h} = \frac{6}{14} = \frac{3}{7},$$

so

$$r = \frac{3}{7}h.$$

The water volume is

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{3}{7}h\right)^2 h = \frac{3\pi}{49}h^3.$$

Differentiate:

$$\frac{dV}{dt} = \frac{9\pi}{49}h^2\frac{dh}{dt}.$$

At the given moment, $h = 4$ and $\dfrac{dh}{dt} = -2$, so

$$\frac{dV}{dt} = \frac{9\pi}{49}(16)(-2) = -\frac{288\pi}{49}.$$

Thus water is leaking out at the rate

$$\boxed{\frac{288\pi}{49}\ \text{ft}^3/\text{min}}.$$

